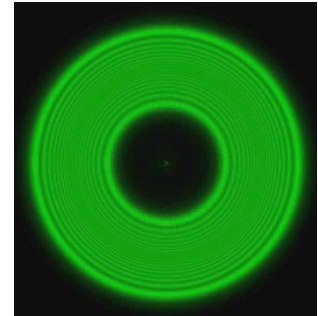
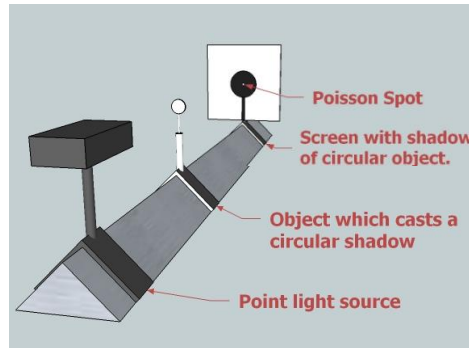


Chap.36 Diffraction



Augustin-Jean Fresnel (1788 –1827) was a French civil engineer and physicist. His research enables the wide acceptance of wave theory of light.

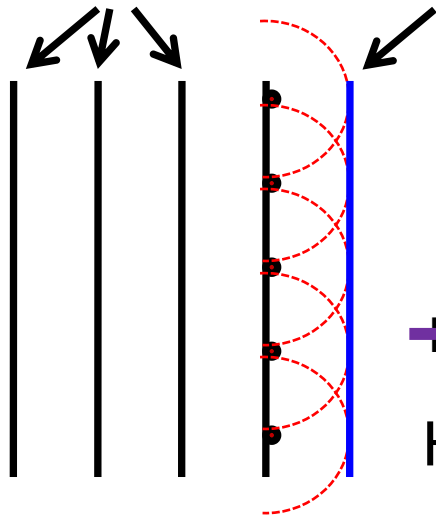


I work far less to capture the public's votes than to obtain an inner approbation which has always been the sweetest reward of my efforts...but all the compliments I received from MM. Arago, Laplace, and Biot never gave me as much pleasure as the discovery of a theoretical truth and the confirmation of my calculations by experiment.

--- 1824 Letter to Young

Wave nature of light

wavefronts new wavefront

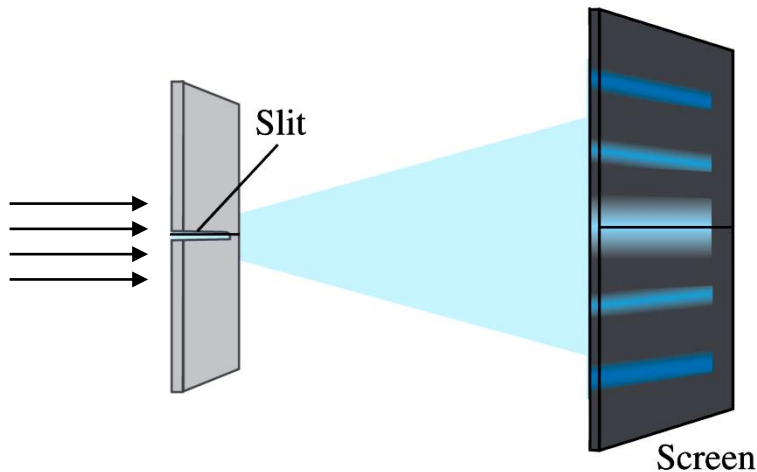
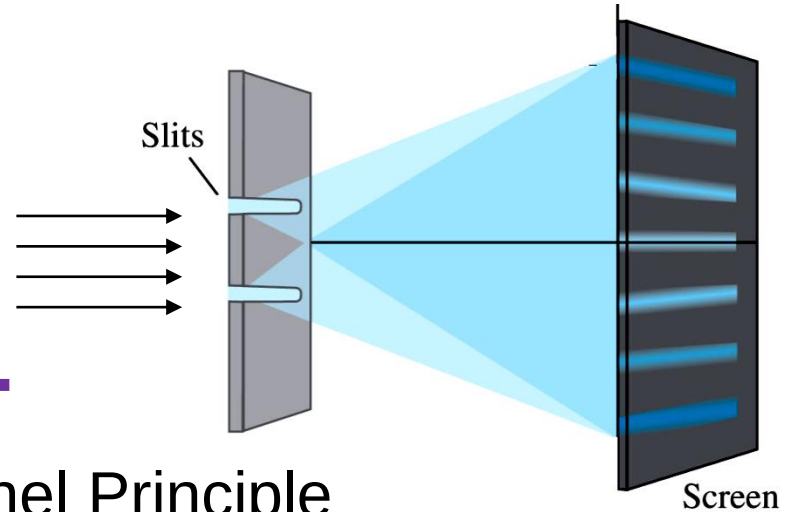


ray



~~Huygens' Principle~~

Huygens-Fresnel Principle



diffraction

Young's double-slit experiment

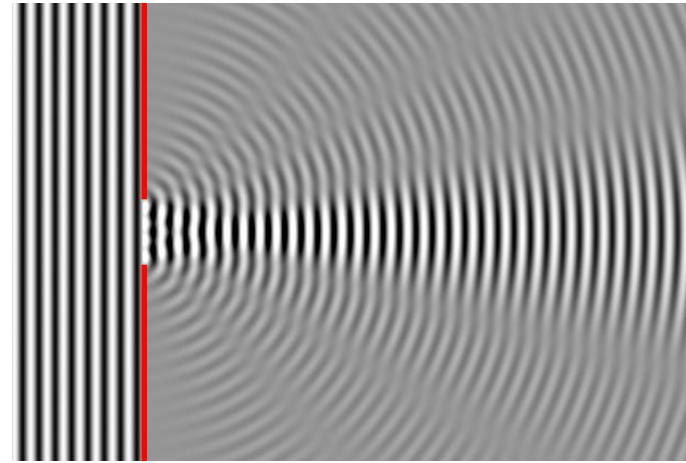
Diffraction is not limited to narrow opening (**slit or pinhole**), also occurs when light passes an **edge**.

Chapter 36 Diffraction

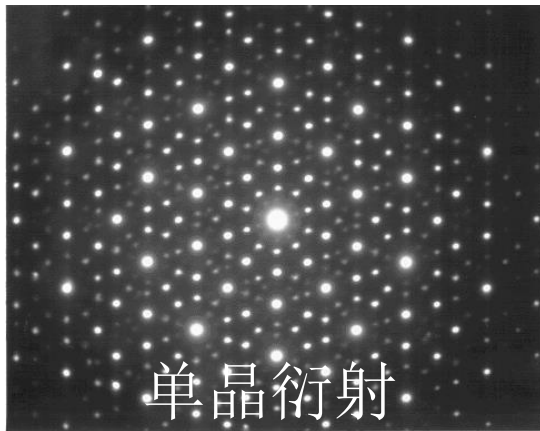
- Single-Slit Diffraction
- Intensity In Single-Slit Diffraction
- Diffraction By A Circular Aperture
- Diffraction By A Double Slit
- Diffraction Gratings *
- Gratings: Dispersion & Resolving Power *
- X-ray Diffraction *

<https://www.bilibili.com/video/av200438183/>

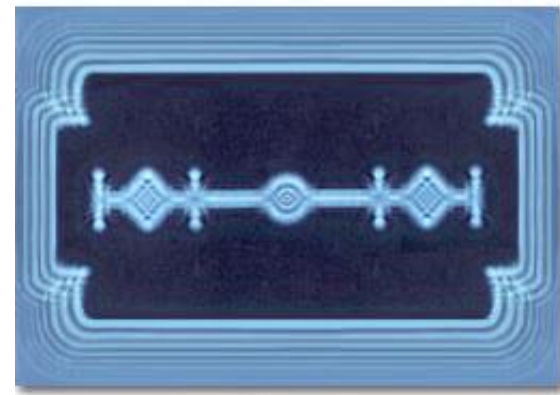
Diffraction of waves



波绕过障碍物（故衍射 也称“绕射”） 波穿过小孔



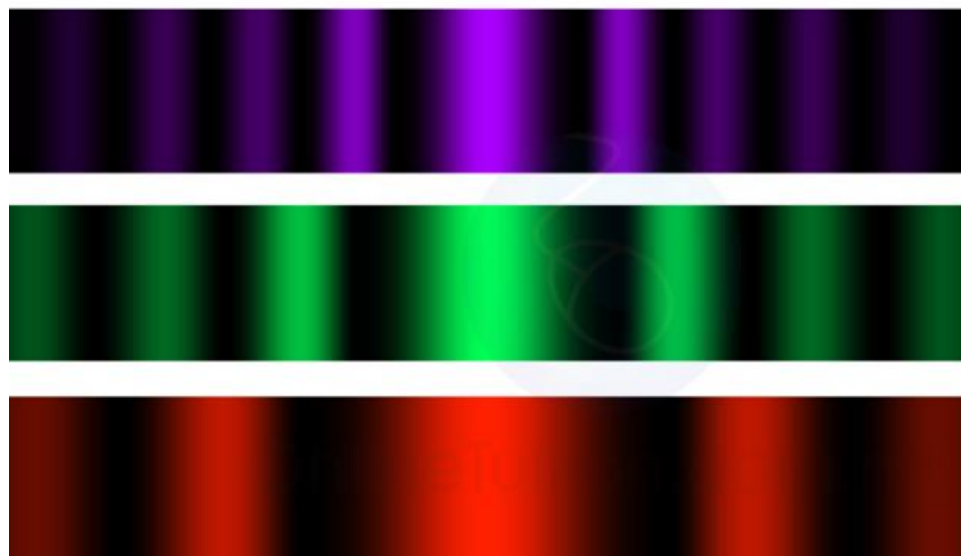
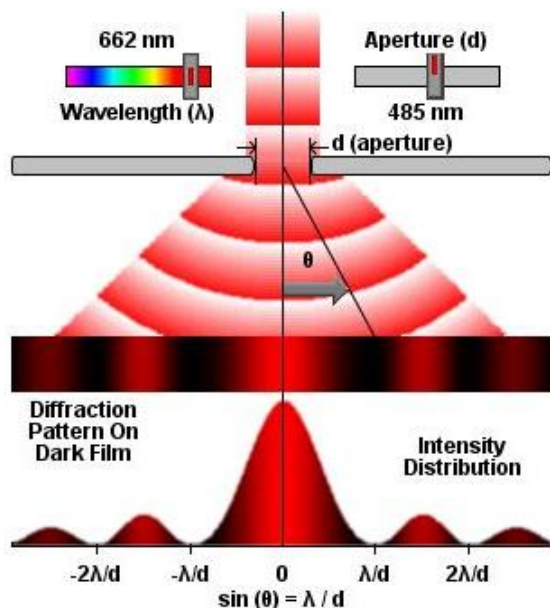
单晶衍射



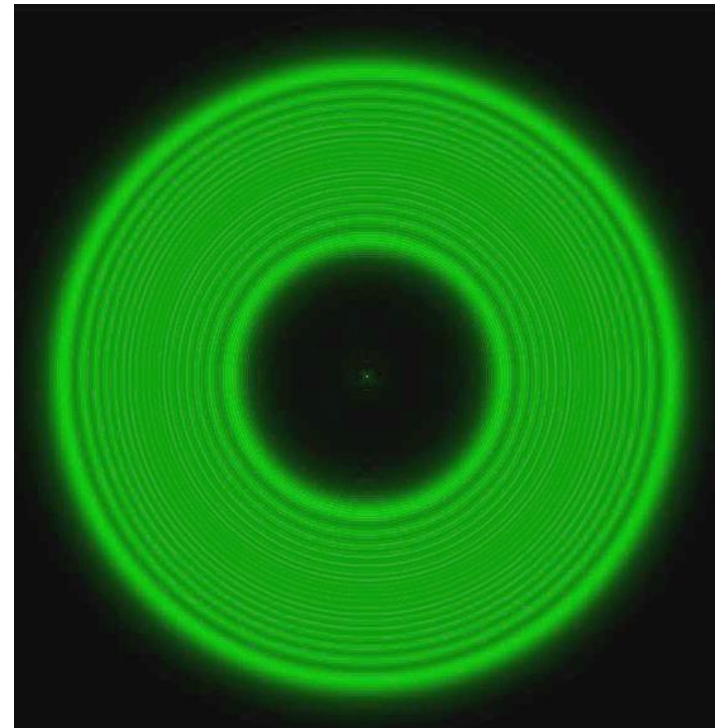
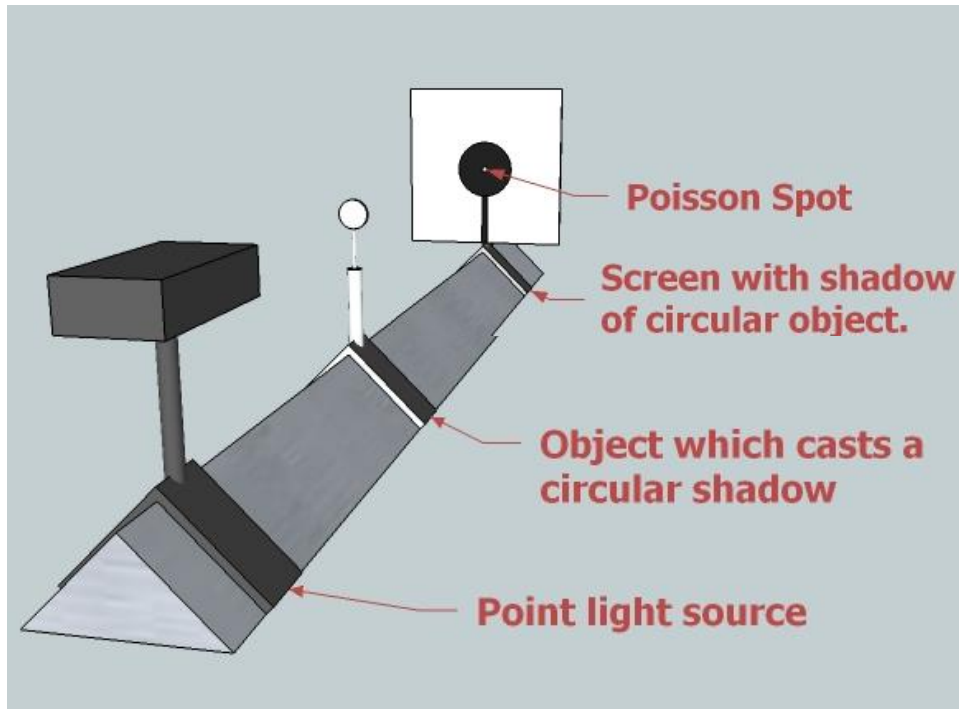
衍射(绕射) + 干涉

Light Diffraction (光的衍射)

Diffraction (衍射) and interference (干涉) describes basically the same phenomenon of light. When monochromatic light passes through a narrow slit and is then intercepted by a viewing screen, the light produces on the screen a diffraction pattern consisting of a broad and intense **central maximum** plus a number of narrower and less intense maxima (**secondary** or **side** maxima) to both sides.



Fresnel(菲涅耳) bright spot / Arago spot / Poisson spot (1819)



Siméon Denis Poisson

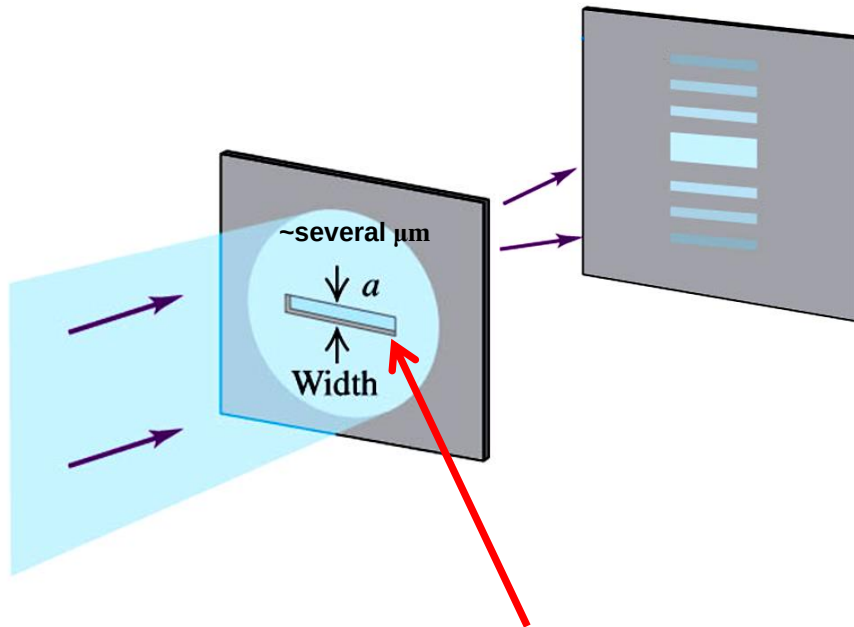


François Arago

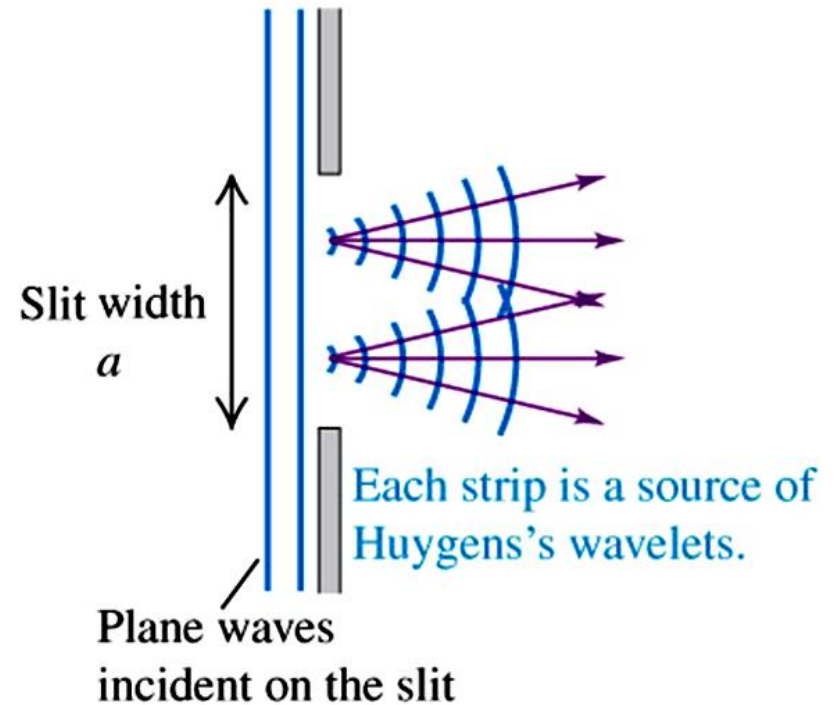


Augustin-Jean Fresnel

Diffraction

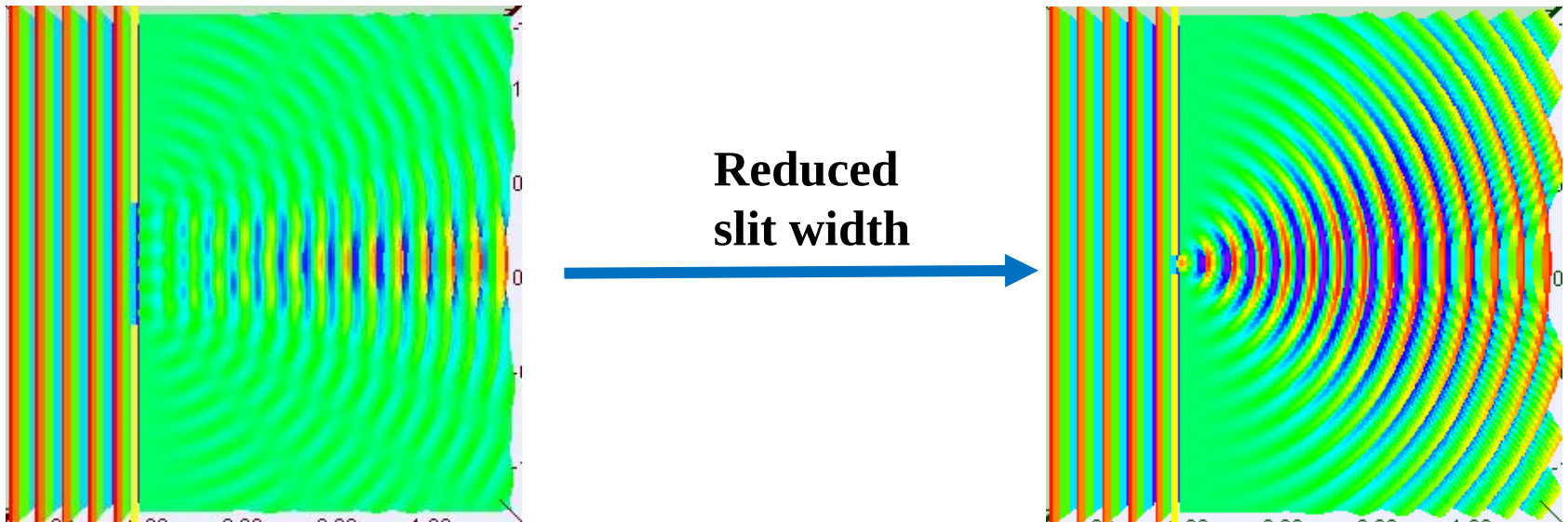


Here, by Huygen's principle, we consider every point along the width of the slit as a point source



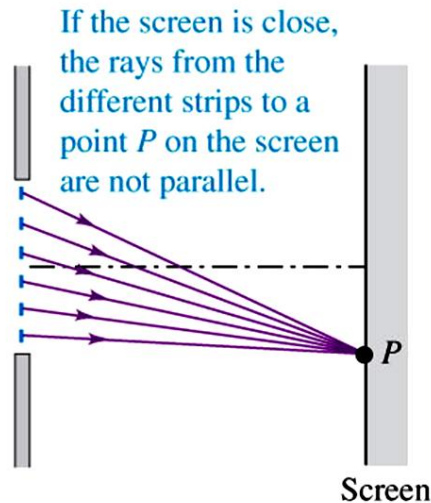
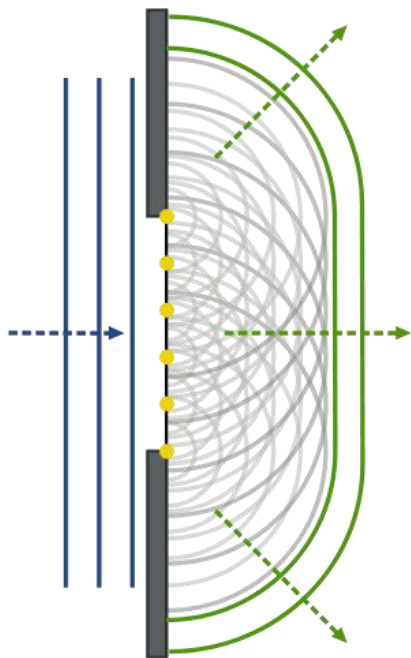
Diffraction by a single slit

A long slit of infinitesimal width which is illuminated by light diffracts the light into a series of circular waves. The width of the slit, as compared with the wavelength, is crucial to the diffraction pattern. A slit which is wider than a wavelength produces interference effects in the space downstream of the slit, while a slit whose width is comparable to the wavelength “rounds” the light more significantly into the shadow regime.

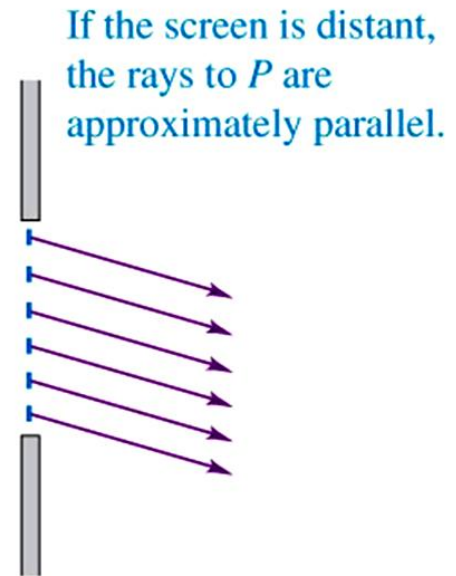


Fresnel and Fraunhofer diffraction

For the study of slit diffraction, every point in the slit can be considered as point sources producing in-phase wavefronts (Huygens' principle). Light rays are not parallel for a point P close to the slit (**Fresnel diffraction**), while they are for a far-away point (**Fraunhofer diffraction**).

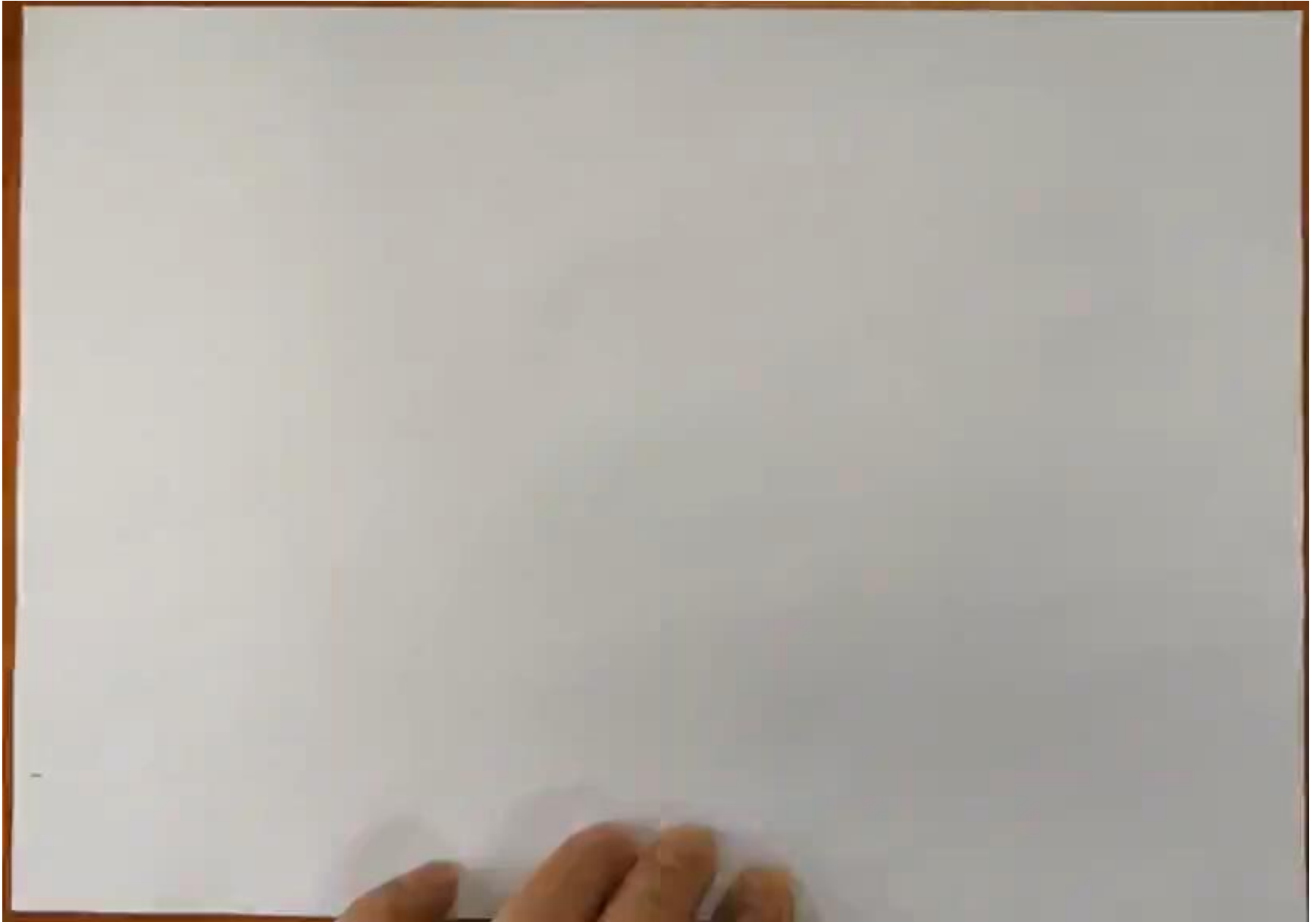


Fresnel (near-field) diffraction
(菲涅耳)

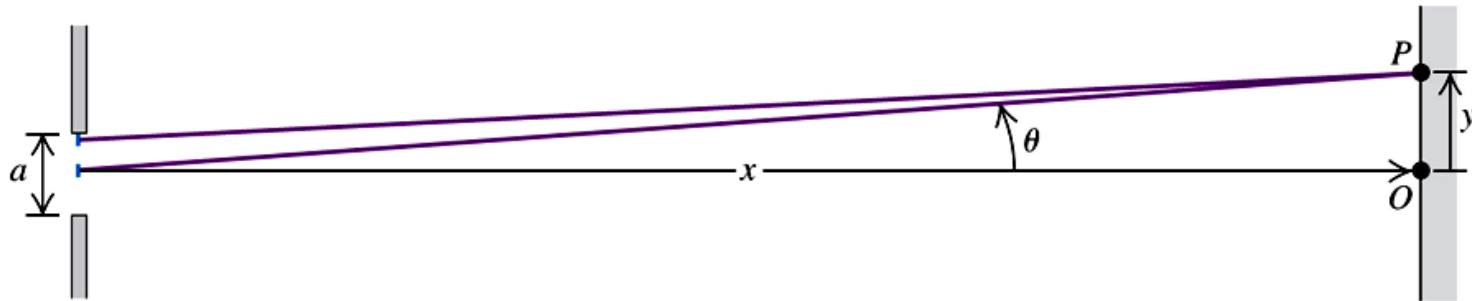


Fraunhofer (far-field) diffraction
(夫琅和费)

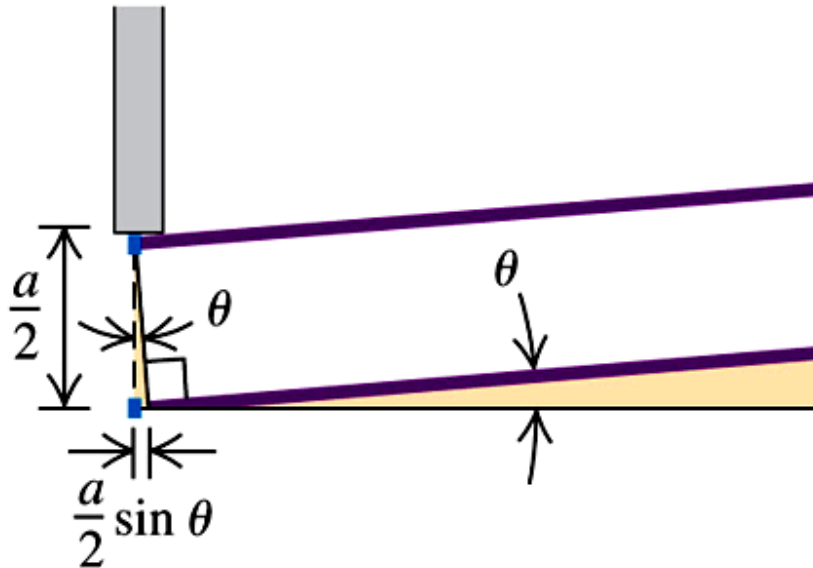
Fraunhofer (far-field) diffraction



Diffraction



Enlarged view of the top half of the slit



A dark fringe occurs whenever :

$$\frac{a}{2} \sin \theta = \pm \frac{\lambda}{2}$$

With this argument, if we divide the slit into 4, 6, 8.....etc, we can calculate the angle of more dark fringes :

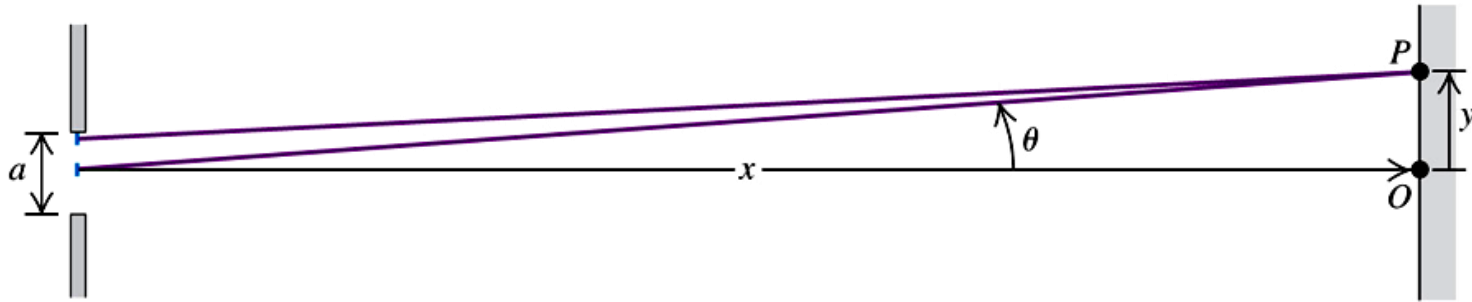
$$\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots)$$

**$m = 0$ is not
a minimum!
Why?**

Since **$\sin \theta \approx \theta$** when **$\theta$** is small

$$\theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \quad (\text{for small angles } \theta)$$

Position of dark fringes on screen



Dark fringes:

$$\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots)$$

$$\theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \\ \text{(for small angles } \theta \text{)}$$

$$\tan \theta = y_m/x \quad \tan \theta \approx \theta \quad \text{for small } \theta$$

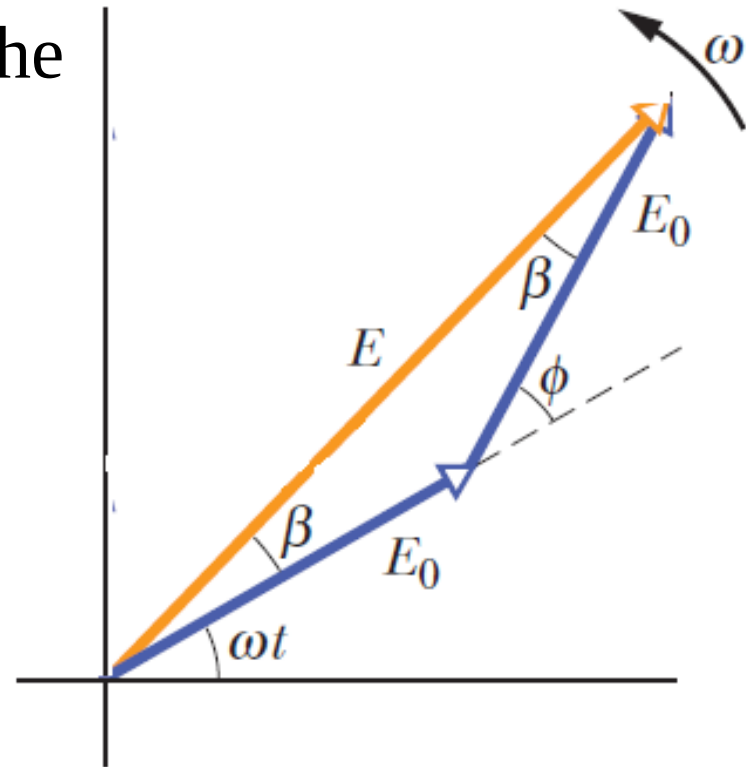
$$y_m/x = \frac{m\lambda}{a} \quad y_m = x \frac{m\lambda}{a}$$

Finding the intensity of the interference pattern

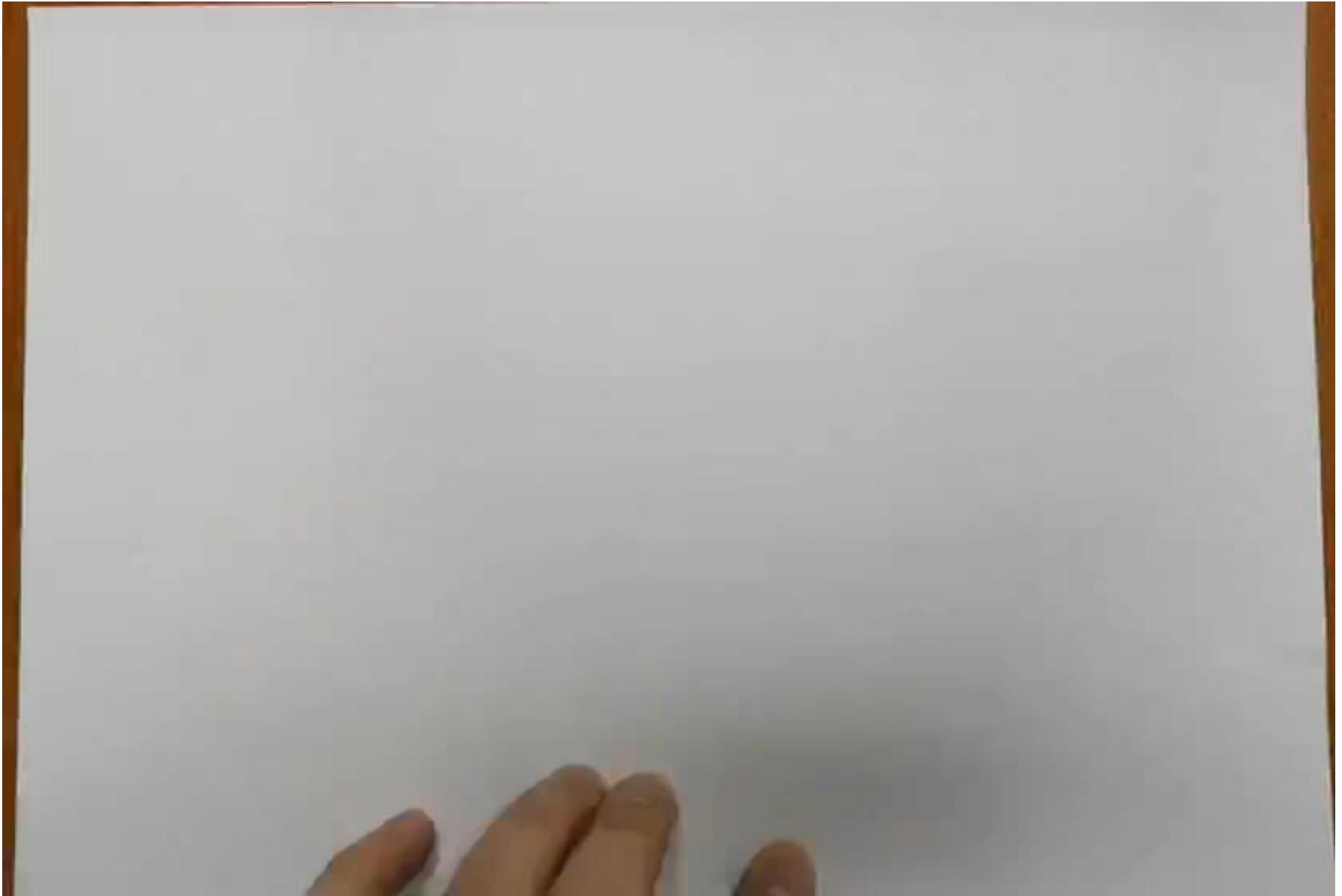
Young's Interference experiment

We know $I \propto E^2$ (intensity is proportional to the square of the amplitude)

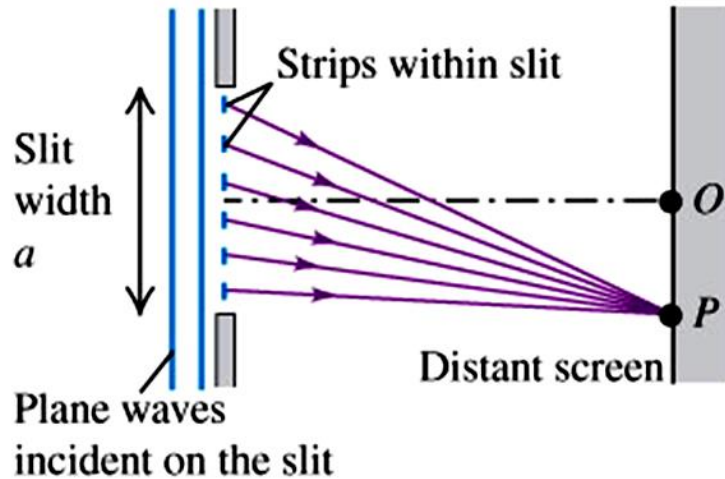
$$I = 4I_0 \cos^2 \frac{1}{2} \phi$$



Intensity in single-slit diffraction



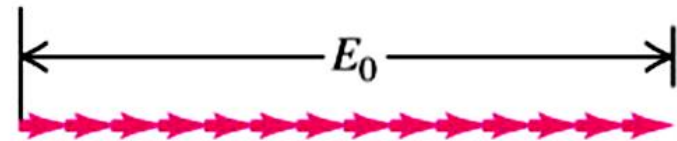
Intensity of the single-slit diffraction pattern



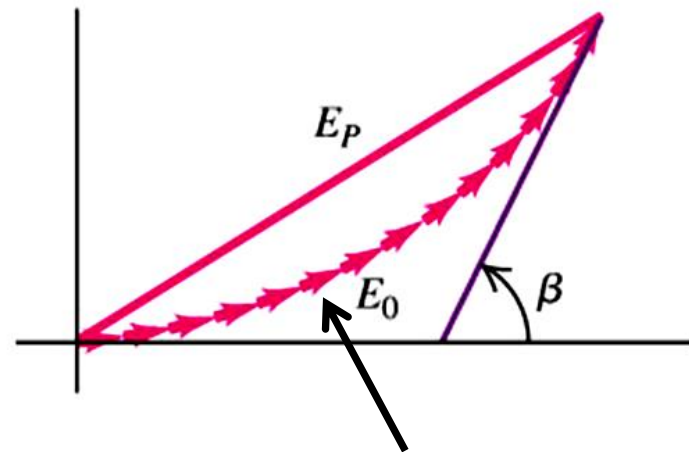
According to Huygens' Principle, to obtain the E-field at any given point, say P , we should consider the contribution of all point sources.

To do so, we can divide the slit into many tiny strips, and the contribution to the total E-field at any given point by each strip can be represented by a tiny vector, so the total E-field is the sum of all the tiny vectors.

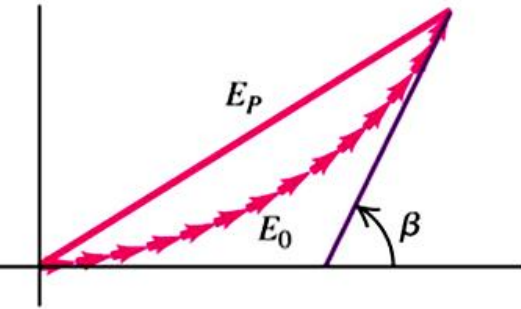
At the center of the diffraction pattern (point O), the phasors are in phase.



Phasor diagram at a point slightly off the center of the pattern; β = total phase difference between the first and last phasors.



The length of this arc is still E_0 !!

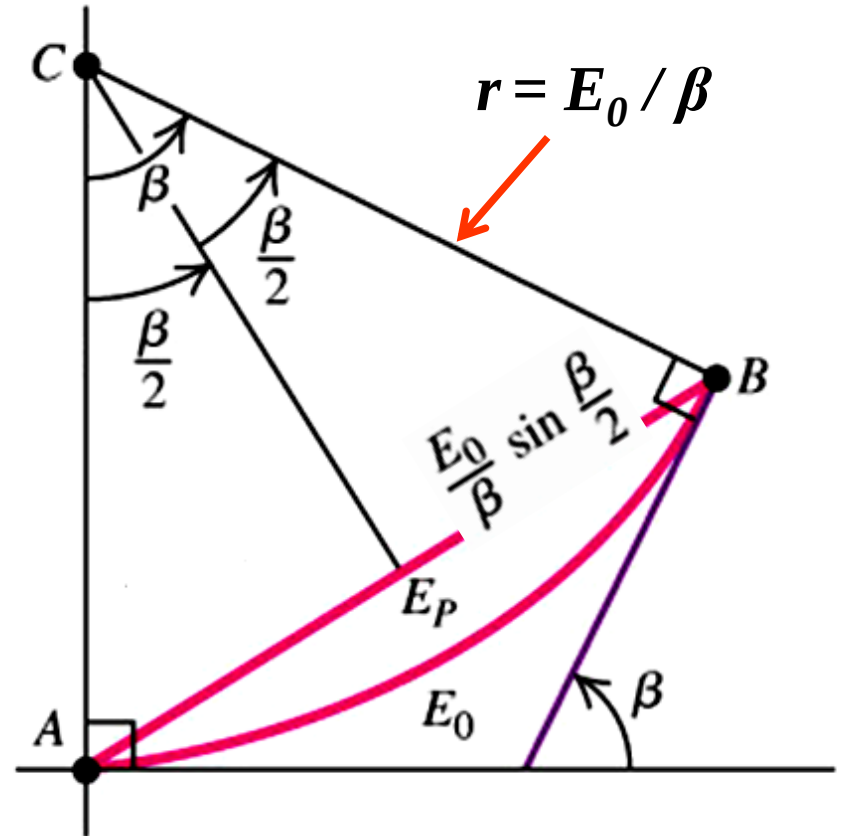


β : phase difference in radians between the first and last point source at the single slit

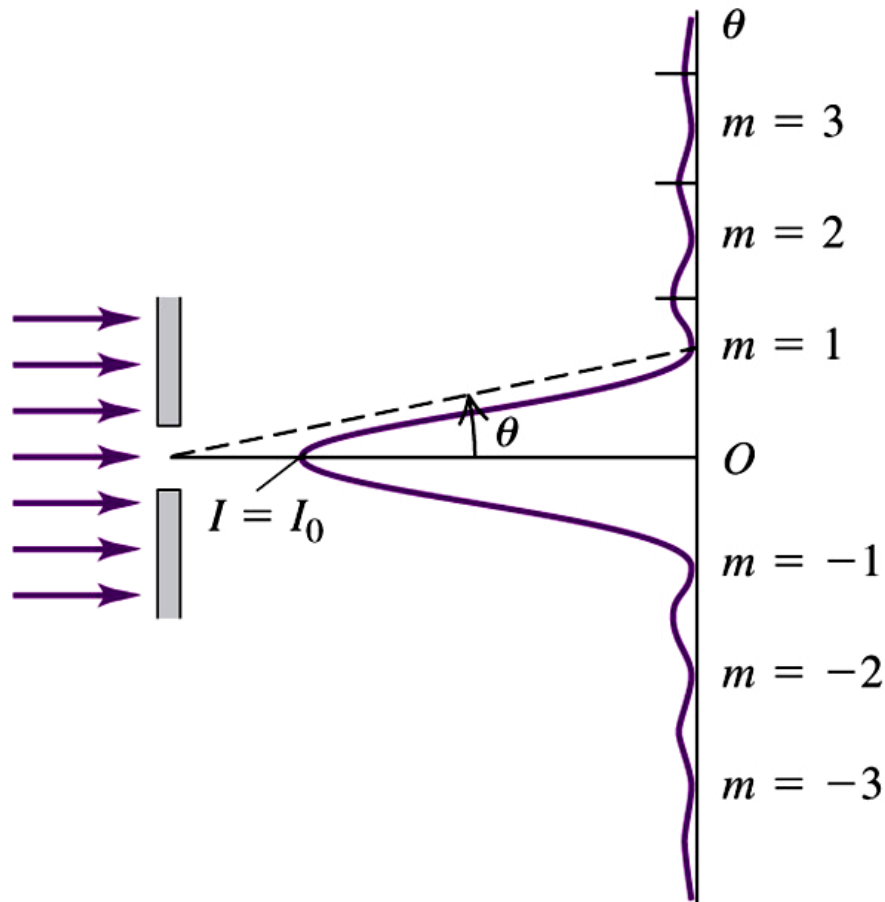
$$E_p = 2 \frac{E_0}{\beta} \sin \frac{\beta}{2}$$

$$= E_0 \frac{\sin(\beta/2)}{\beta/2}$$

$$I = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2$$



Intensity of the single-slit diffraction pattern



Angular position θ of the dark fringes :

$$\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots)$$

Intensity at any point on the screen in terms of phase difference β for the first and last point source at the slit :

$$I = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2$$

To relate the two, we have

$$\beta = \frac{2\pi}{\lambda} a \sin \theta$$

Example

- (a) The intensity at the center of a single-slit diffraction pattern is I_0 . What is the **intensity** at a point in the pattern where there is a **66-radian phase difference** between wavelets from the two edges of the slit?
- (b) If this point is **7° away** from the central maximum, **how many wavelengths wide is the slit?**

$$I = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2$$

$$\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots)$$

$$\sin \theta = \frac{\beta}{2\pi} \frac{\lambda}{a}$$

(a) We have $\beta/2 = 33 \text{ rad}$,

$$\begin{aligned} I &= I_0 \left[\frac{\sin(33 \text{ rad})}{33 \text{ rad}} \right]^2 \\ &= (9.2 \times 10^{-4}) I_0 \end{aligned}$$

(b)

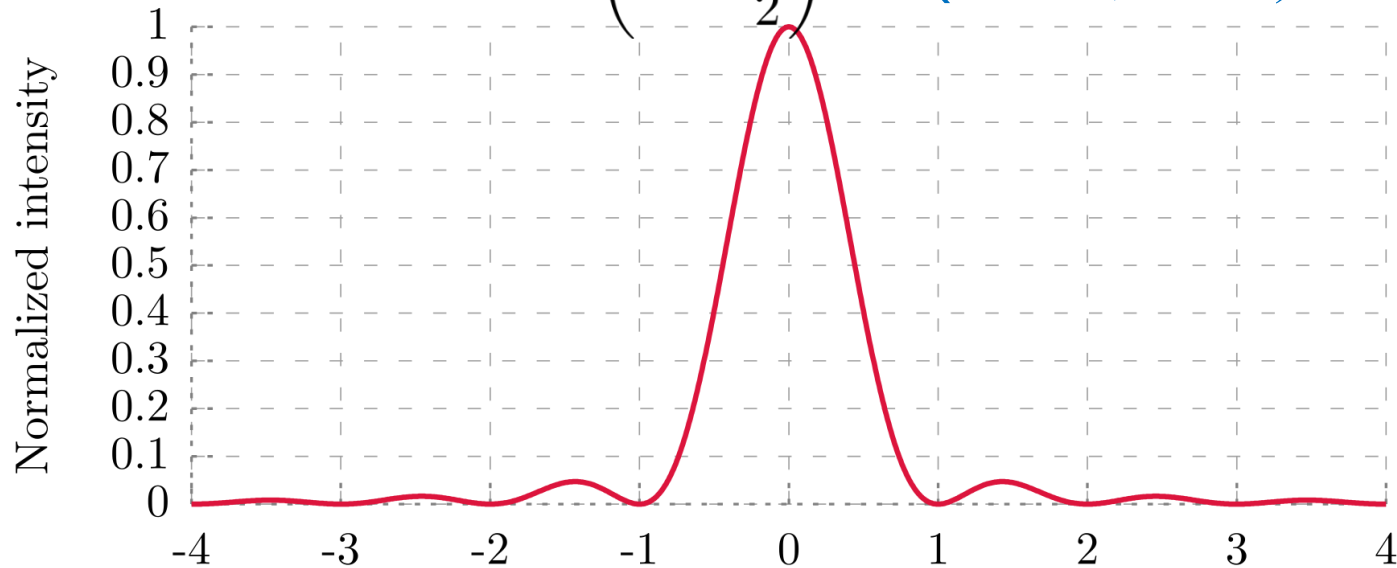
$$\begin{aligned} \frac{a}{\lambda} &= \frac{\beta}{2\pi \sin \theta} = \frac{66 \text{ rad}}{(2\pi \text{ rad}) \sin 7.0^\circ} \\ &= 86 \end{aligned}$$

Example

$$\Rightarrow I_{\theta} = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2.$$

Find the intensities of the first three secondary maxima (side maxima) in the single-slit diffraction pattern, measured as a percentage of the intensity of the central maximum. Assume that these side maxima occurs at

$$\alpha = \left(m + \frac{1}{2} \right) \pi. \quad (m=+1, +2\dots)$$



$$a \sin \theta = m\lambda. \quad \text{--- -- -- -- --} \rightarrow \quad a \cdot \sin(\theta)/\lambda$$

Example

Solution:

The answer to this question is very straightforward. Just use

$$\frac{I_{\theta}}{I_m} = \left(\frac{\sin \alpha}{\alpha} \right)^2 = \left(\frac{\sin (m + 1/2) \pi}{(m + 1/2) \pi} \right)^2 ,$$

we have

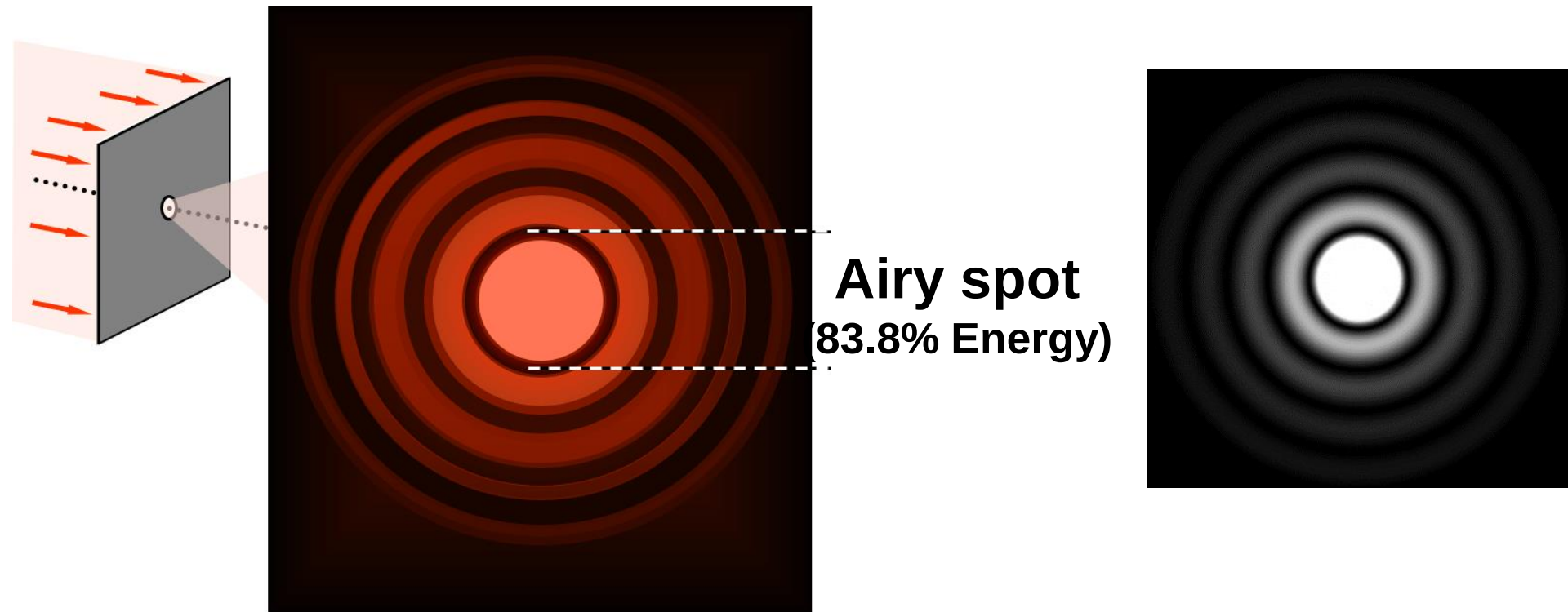
$$\frac{I_1}{I_m} = \left(\frac{\sin (1 + 1/2) \pi}{(1 + 1/2) \pi} \right)^2 = \left(\frac{\sin 1.5\pi}{1.5\pi} \right)^2 = 4.5\% ,$$

$$\frac{I_2}{I_m} = \left(\frac{\sin (2 + 1/2) \pi}{(2 + 1/2) \pi} \right)^2 = \left(\frac{\sin 2.5\pi}{2.5\pi} \right)^2 = 1.6\% ,$$

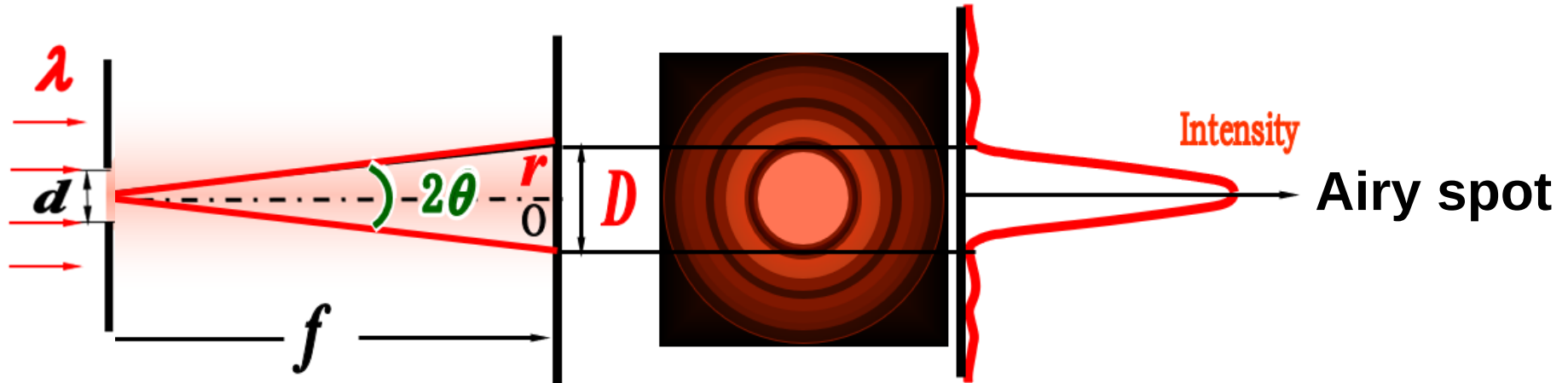
$$\frac{I_3}{I_m} = \left(\frac{\sin (3 + 1/2) \pi}{(3 + 1/2) \pi} \right)^2 = \left(\frac{\sin 3.5\pi}{3.5\pi} \right)^2 = 0.83\% .$$

Diffraction by a circular aperture

If light passes a small circular aperture (hole), the diffraction pattern appears as a circular disk (the **Airy disk**) surrounded by several progressively fainter secondary rings. The intensity distribution is given by



Diffraction by a circular aperture



Intensity of the Fraunhofer diffraction for a circular aperture:

$$I_P = A_0^2 \left[1 - \frac{1}{2}m^2 + \frac{1}{3}\left(\frac{m^2}{2!}\right)^2 - \frac{1}{4}\left(\frac{m^3}{3!}\right)^2 + \frac{1}{5}\left(\frac{m^4}{4!}\right)^2 + \dots \right]^2 \quad m = \frac{\pi d \sin \theta}{\lambda}$$

Based on the 1st order Bessel function, we have

$$I_P = I_0 \left[\frac{J_1(2m)}{m} \right]^2.$$

Diffraction by a circular aperture

$$I_{\theta} = I_m \left[\frac{2J_1 (ka \sin \theta)}{(ka \sin \theta)} \right]^2 ,$$

where $k = 2\pi/\lambda$, a is the radius of the aperture ($d = 2a$), and J_1 is the 1st order Bessel function.

1st Maxima:

$$\sin \theta_0 = 0$$

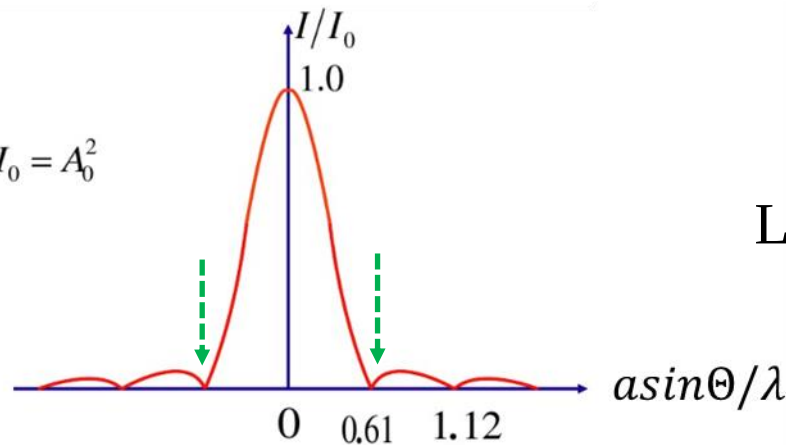
$$I_0 = A_0^2$$

Minima:

$$\sin \theta_1 = 0.610 \frac{\lambda}{a}$$

$$\sin \theta_2 = 1.116 \frac{\lambda}{a}$$

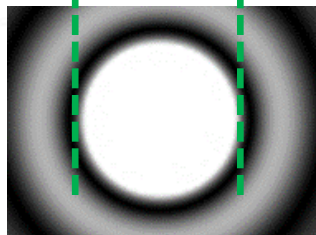
$$\sin \theta_3 = 1.619 \frac{\lambda}{a}$$



Location of the 1st minimum:

$$\sin \theta = 1.22 \frac{\lambda}{d}$$

$$(d = 2a)$$



<https://zhuanlan.zhihu.com/p/47622756>

Diffraction by a circular aperture

Secondary maxima: **The relative intensity of**
Secondary maxima:

$$\sin \theta_{10} = 0.819 \frac{\lambda}{R}$$

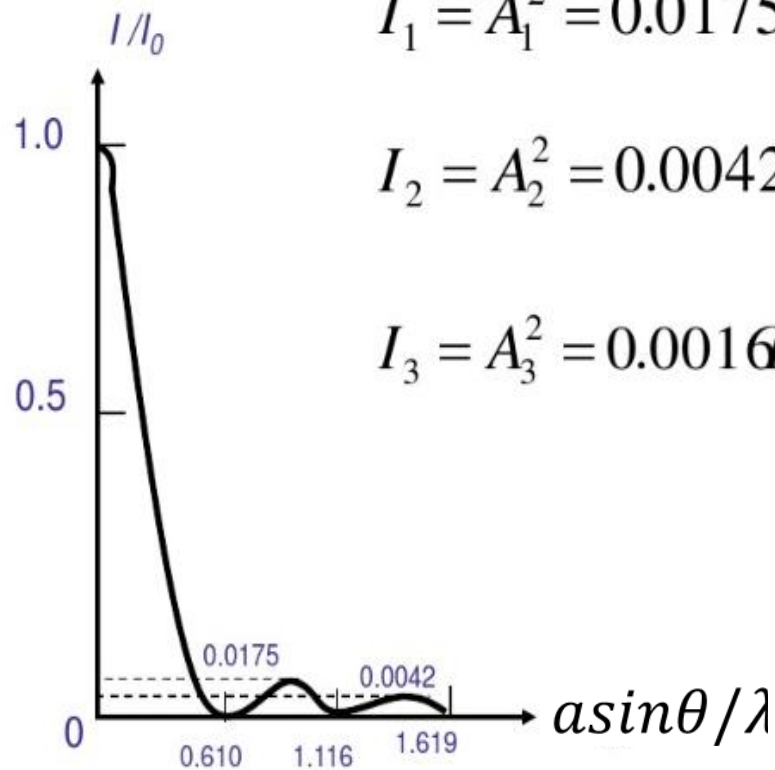
$$\sin \theta_{20} = 1.333 \frac{\lambda}{R}$$

$$\sin \theta_{30} = 1.847 \frac{\lambda}{R}$$

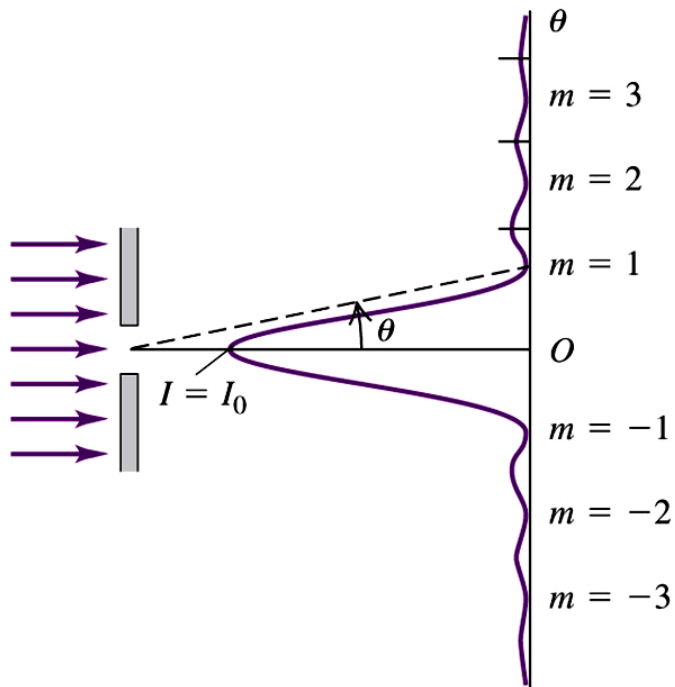
$$I_1 = A_1^2 = 0.0175 I_0$$

$$I_2 = A_2^2 = 0.0042 I_0$$

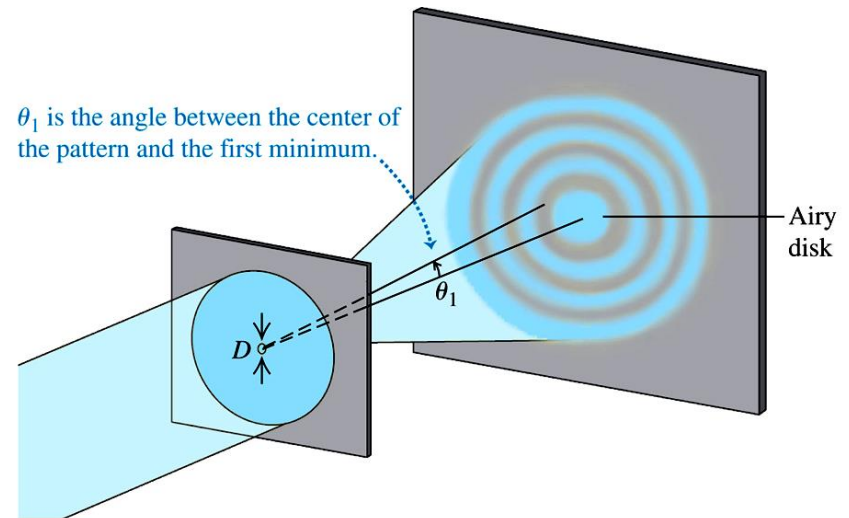
$$I_3 = A_3^2 = 0.0016 I_0$$



Diffraction by a circular aperture and resolvability



$$\sin \theta = \frac{\lambda}{a} \quad (\text{first minimum—single slit}).$$



For circular aperture

$$\sin \theta = 1.22 \frac{\lambda}{D} \quad (\text{first minimum—circular aperture})$$

next two dark rings

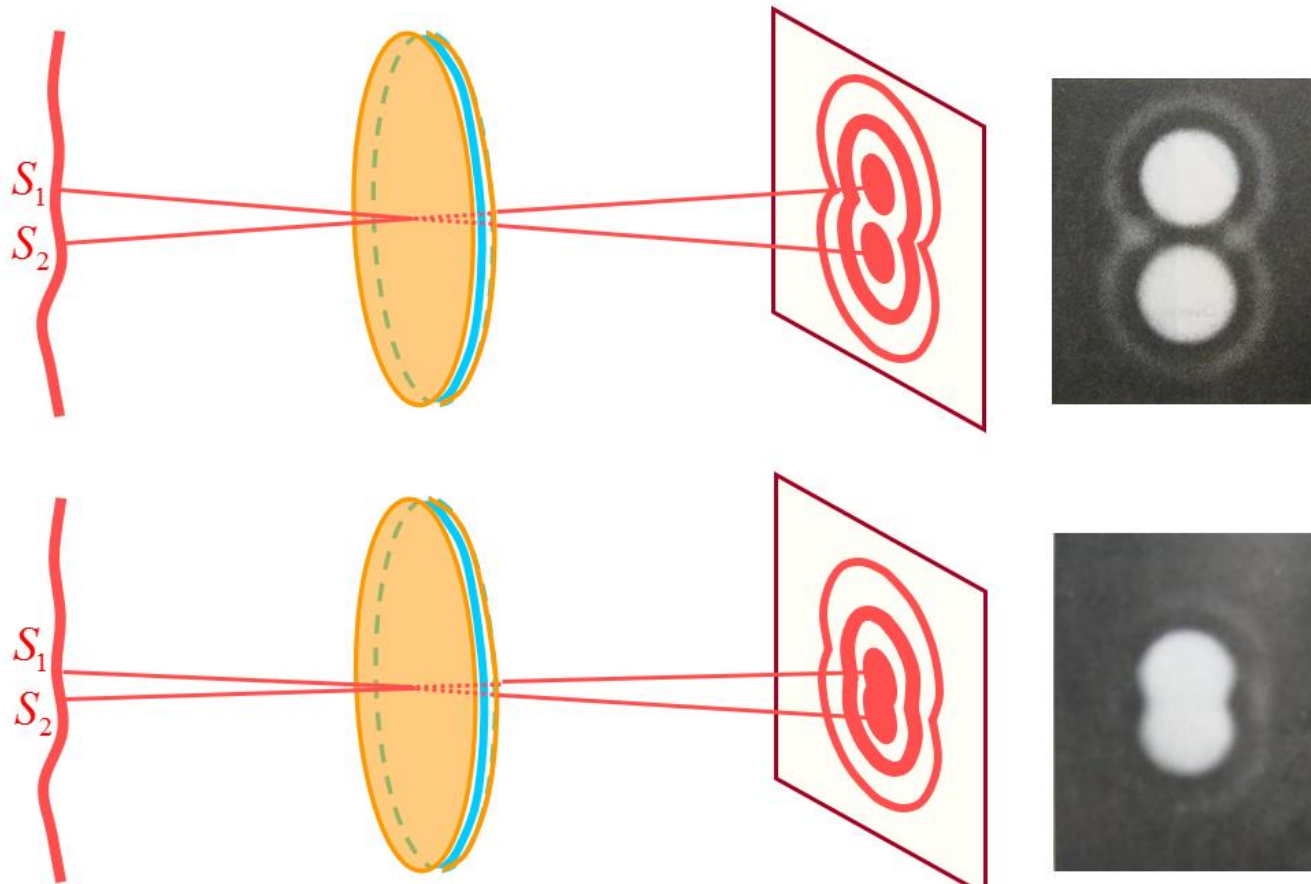
$$\sin \theta_2 = 2.23 \frac{\lambda}{D} \quad \sin \theta_3 = 3.24 \frac{\lambda}{D}$$

bright rings

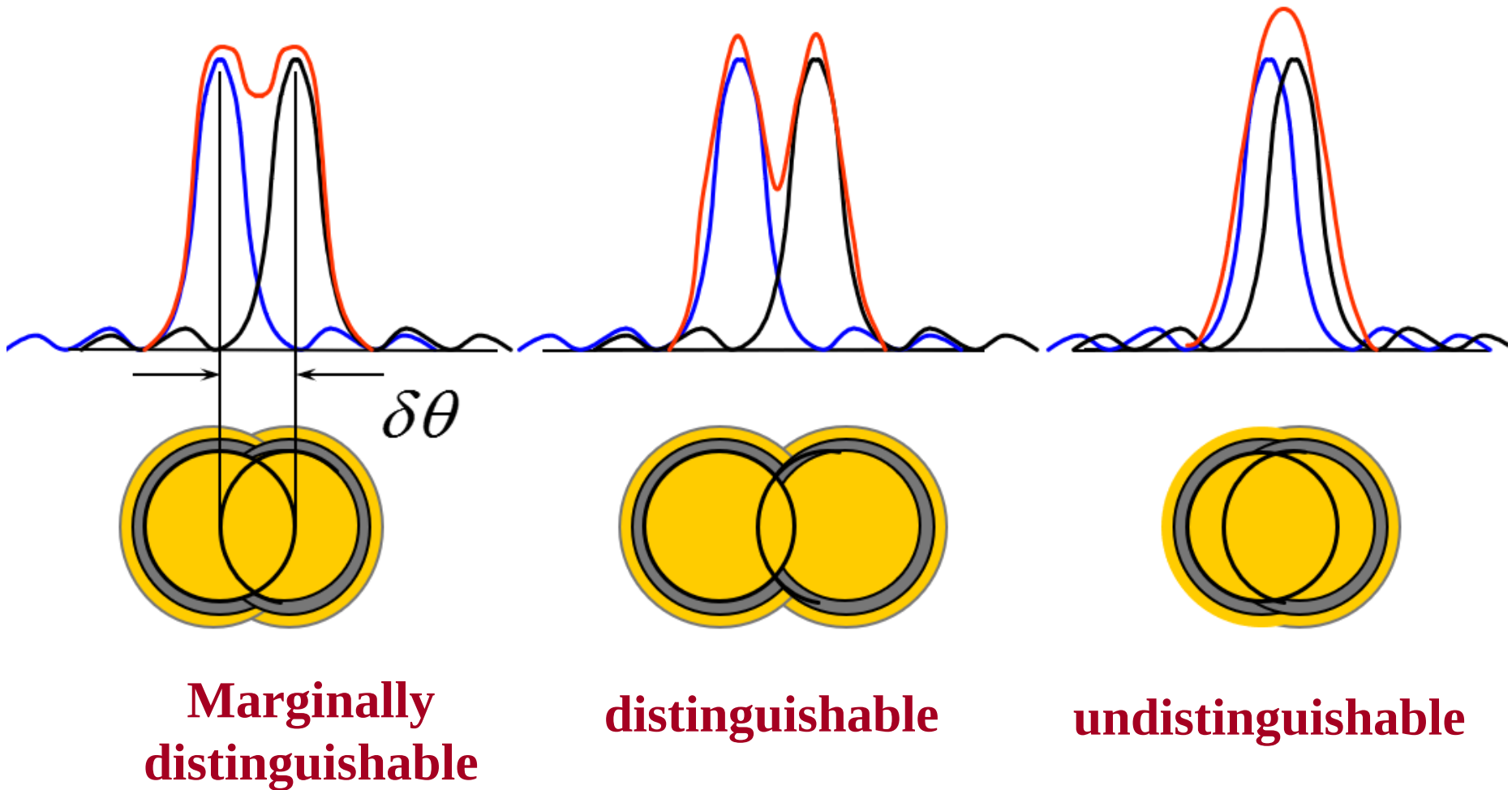
$$\sin \theta = 1.63 \frac{\lambda}{D}, \quad 2.68 \frac{\lambda}{D}, \quad 3.70 \frac{\lambda}{D}$$

Application

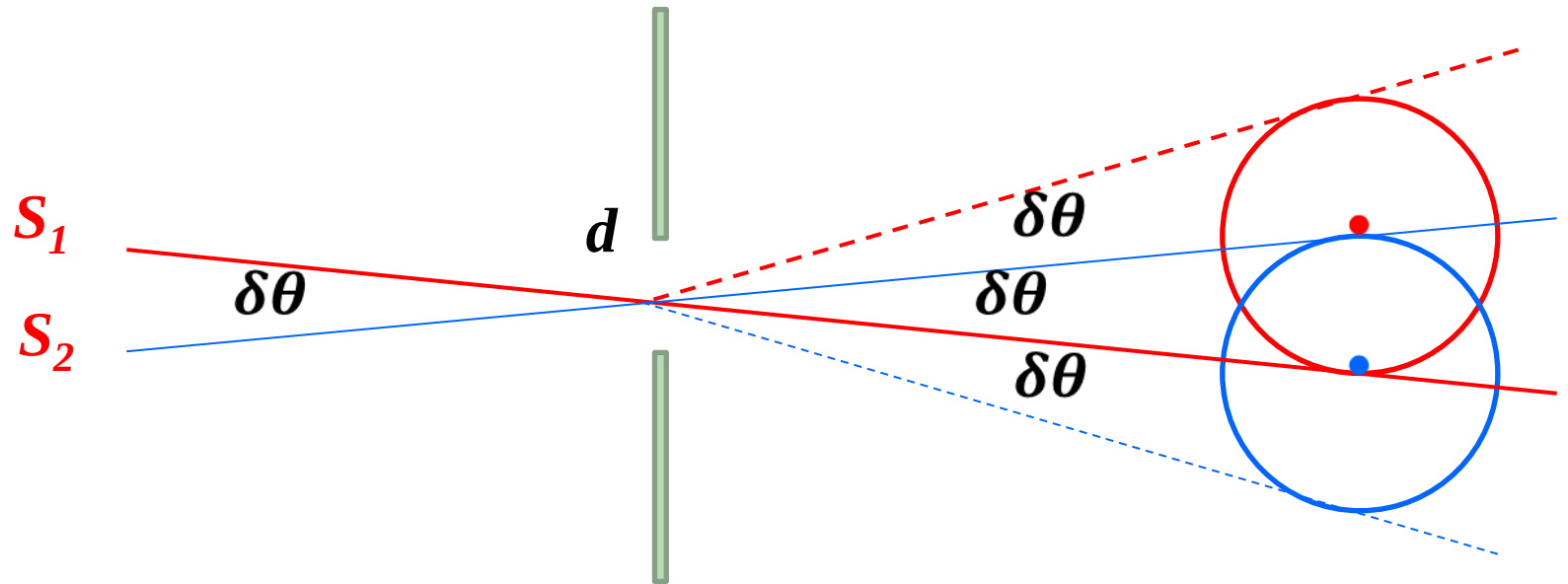
Rayleigh's criterion for resolvability : central maximum of the diffraction pattern of one source is centered on the first minimum of the diffraction pattern of the other



Resolvability



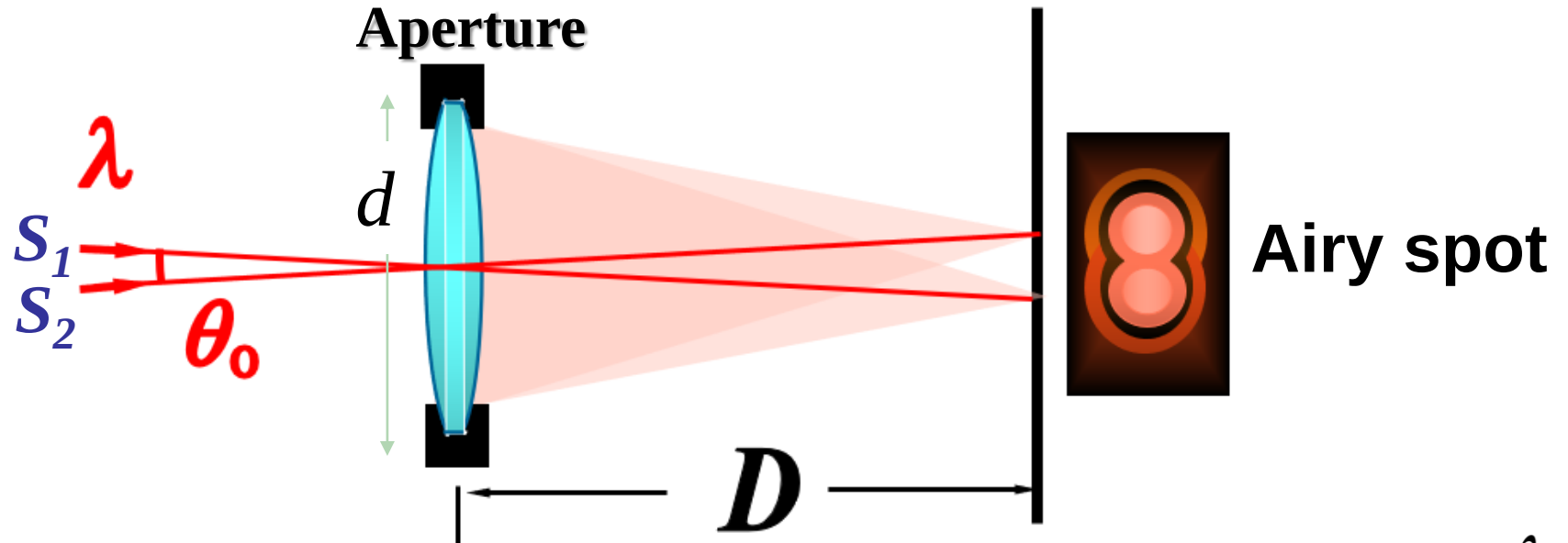
Marginally distinguishable



$$\delta\theta = \theta_0 \approx \sin \theta = 1.22 \frac{\lambda}{d}$$

Rayleigh's criterion

For the **marginally distinguishable** case:

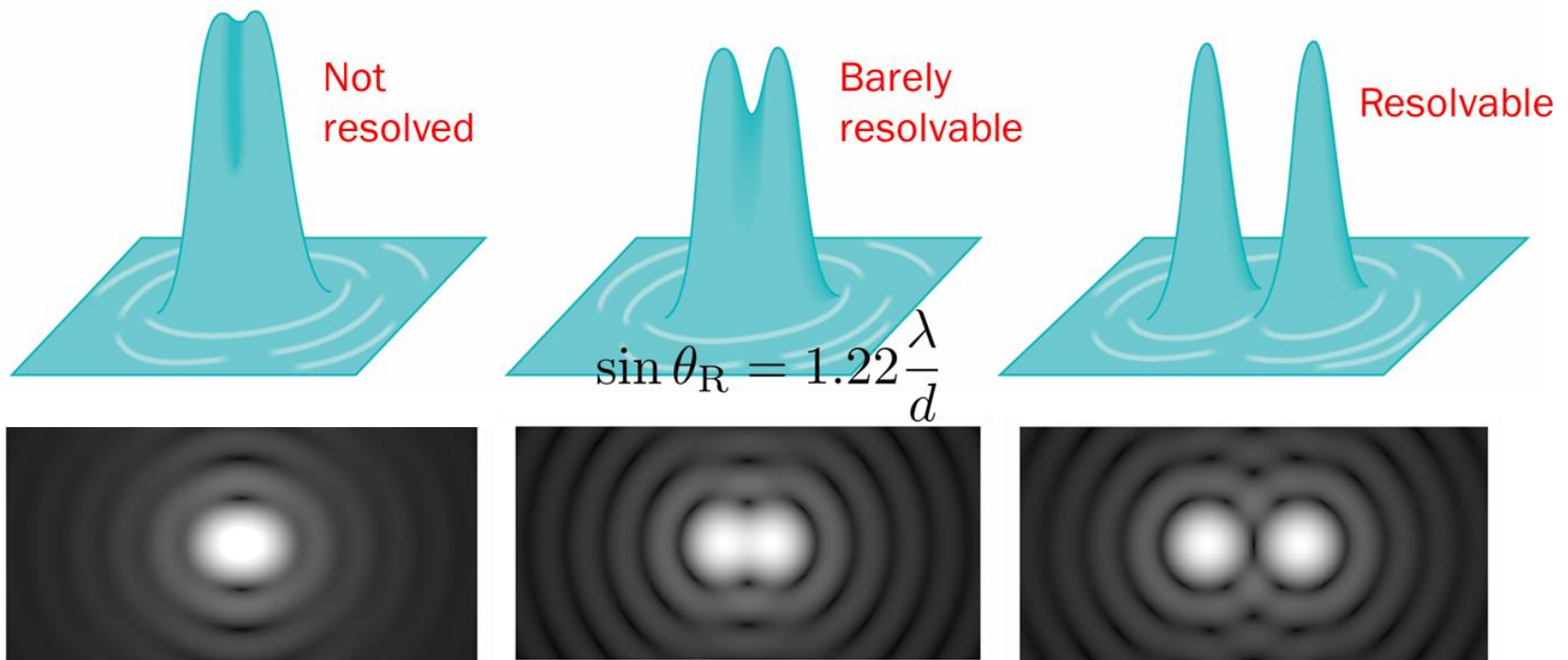


Minimum resolution angle: $\delta\theta = \theta_0 \approx \sin \theta = 1.22 \frac{\lambda}{d}$

Resolution: $R \equiv \frac{1}{\delta\theta} = \frac{d}{1.22\lambda} \left. \begin{matrix} d \uparrow \\ \lambda \downarrow \end{matrix} \right\} \rightarrow R \uparrow$

Resolvability and Rayleigh's criterion

If two point sources move closer, the patterns will start to overlap, and merge to form a single pattern, in which case they cannot be *resolved* in the image. The **Rayleigh criterion** specifies that two point sources can be considered to be resolvable if the separation of the two images is at least the radius of the Airy disk ($\theta_R \approx \sin \theta_R = 1.22\lambda/d$).



Example

A spy satellite is at an altitude 650 km above the Earth's surface. How large must the satellite's camera lens be so that its resolution is 25 cm? Assume the average wavelength of light of 480 nm.

a) 1.8 m

b) 2.7 m

c) 0.55 m

d) 1.5 m

e) 0.85 m

$$\sin \theta = 1.22 \frac{\lambda}{d} \quad (\text{first minimum—circular aperture})$$

Example

You use a telescope lens to form an image of two closely spaced, distant stars. Which of the following will increase the resolving power?

- ✓ A. Use a filter so that only the blue light from the stars enters the lens.
- B. Use a filter so that only the red light from the stars enters the lens.
- C. Use a lens of smaller diameter.
- D. more than one of the above

$$\sin \theta = 1.22 \frac{\lambda}{d} \quad (\text{first minimum—circular aperture})$$

Resolvability and Rayleigh's criterion

China begins operating world's largest radio telescope

September 25, 2016 by Gillian Wong

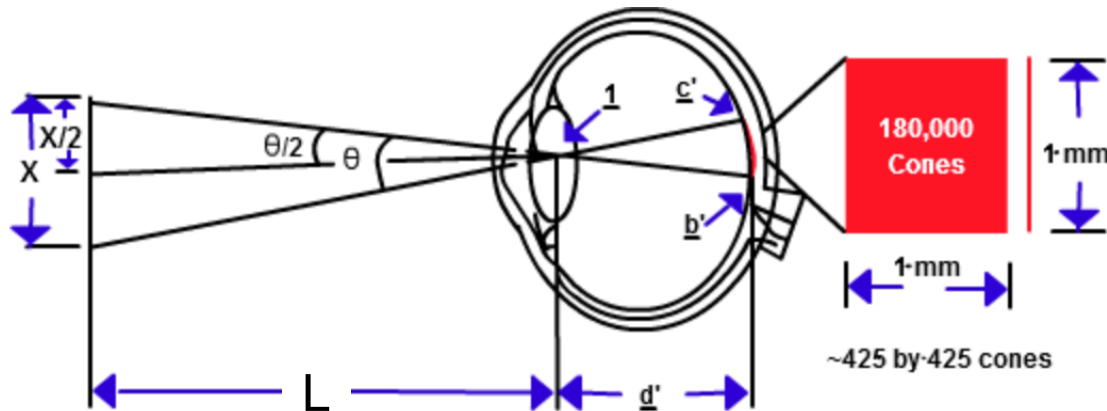


$$R \equiv \frac{1}{\delta\theta} = \frac{d}{1.22\lambda}$$

$$d = 500 \text{ m}$$

In this Saturday, Sept. 24, 2016 photo released by Xinhua News Agency, an aerial view shows the Five-hundred-meter Aperture Spherical Telescope (FAST) in the remote Pingtang county in southwest China's Guizhou province. China has begun operating the world's largest radio telescope to help search for extraterrestrial life. (Liu Xu/Xinhua via AP)

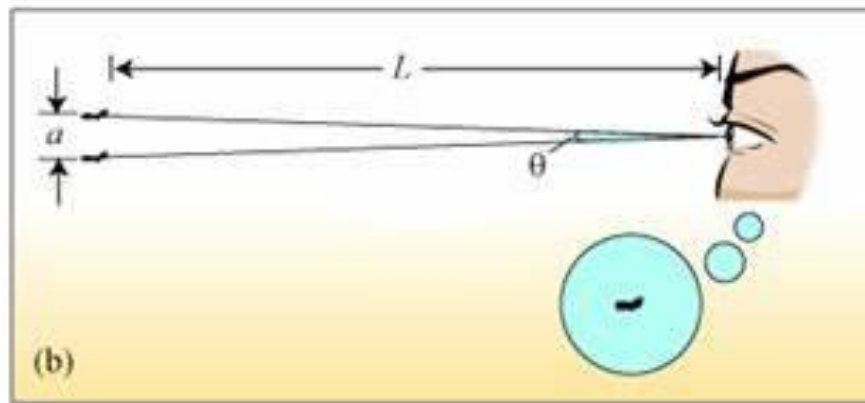
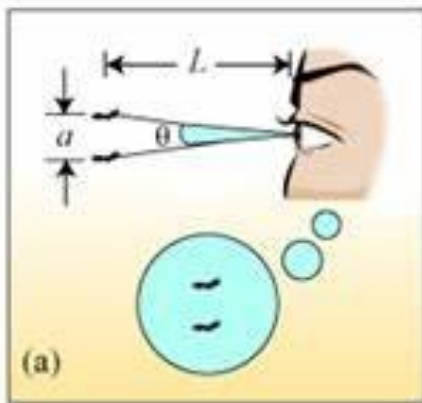
Rayleigh's criterion



Critical vision angle:

$$\delta\theta = \theta_0 \approx \sin \theta = 1.22 \frac{\lambda}{d}$$

$$L \gg x \Rightarrow \theta \approx x/L$$



Rayleigh's criterion

$$\frac{x}{L} = 1.22 \frac{\lambda}{d}$$

Resolvability and Rayleigh's criterion

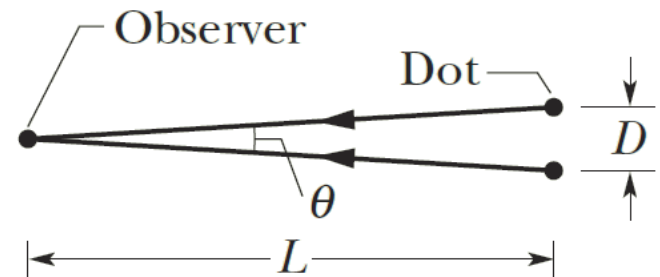


Maximilien Luce, *The Seine at Herblay*, 1890. Musée d'Orsay, Paris, France. Photo by Erich Lessing/Art Resource

Pointillism (点彩画) is a technique of painting in which small, distinct dots of color are applied in patterns to form an image. Pointillism is analogous to the four-color CMYK printing process used by some color printers, and the RGB colors used in televisions and computer monitors.

Example

The figure below is a representation of the colored dots on a pointillistic painting. Assume that the average center-to-center separation of the dots is $D = 2.0$ mm. Also assume that the diameter of the pupil (瞳孔) of your eye is $d = 1.5$ mm and that the least angular separation between dots you can resolve is set only by Rayleigh's criterion. What is the least viewing distance from which you cannot distinguish any dots on the painting?



Example

Solution:

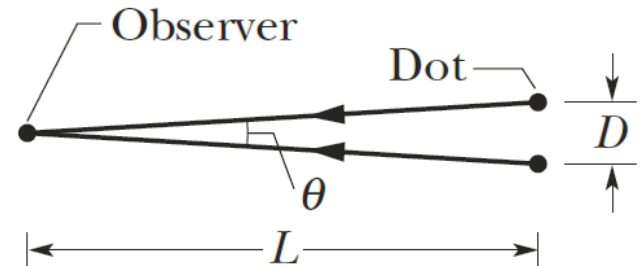
The right figure shows the side view of such setup. Using Rayleigh's criterion and the condition $D \ll L$, we have

$$\theta \approx \frac{D}{L} \approx 1.22 \frac{\lambda}{d} \quad \Rightarrow \quad L = \frac{Dd}{1.22\lambda}.$$

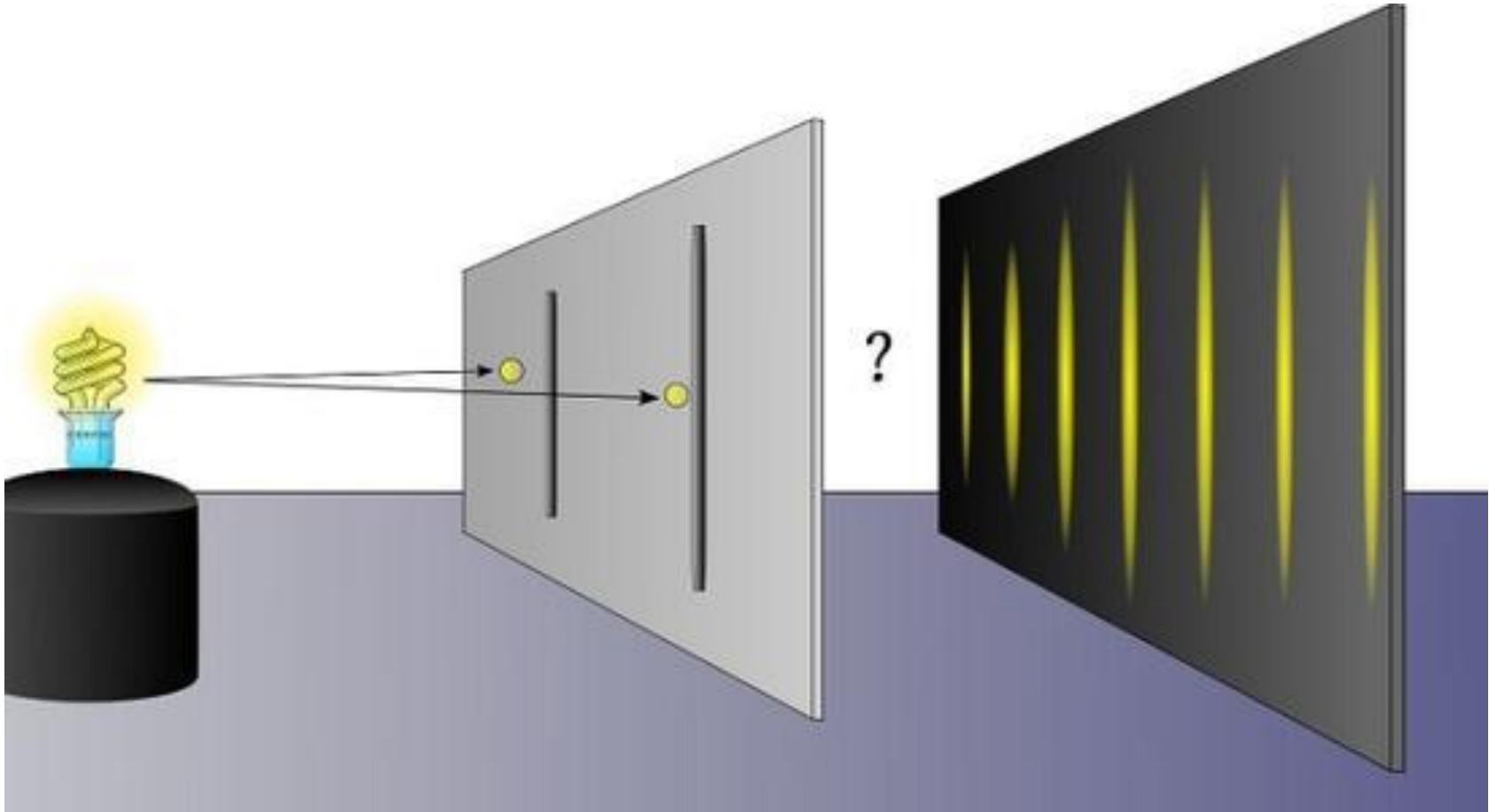
Plug in numbers, we get

$$L = \frac{(2.0 \text{ mm})(1.5 \text{ mm})}{(1.22)(400 \text{ nm})} = 6.1 \text{ m}.$$

At this distance, no visible-colored dots are resolvable, even if the dots are purple (with wave-length 400 nm).

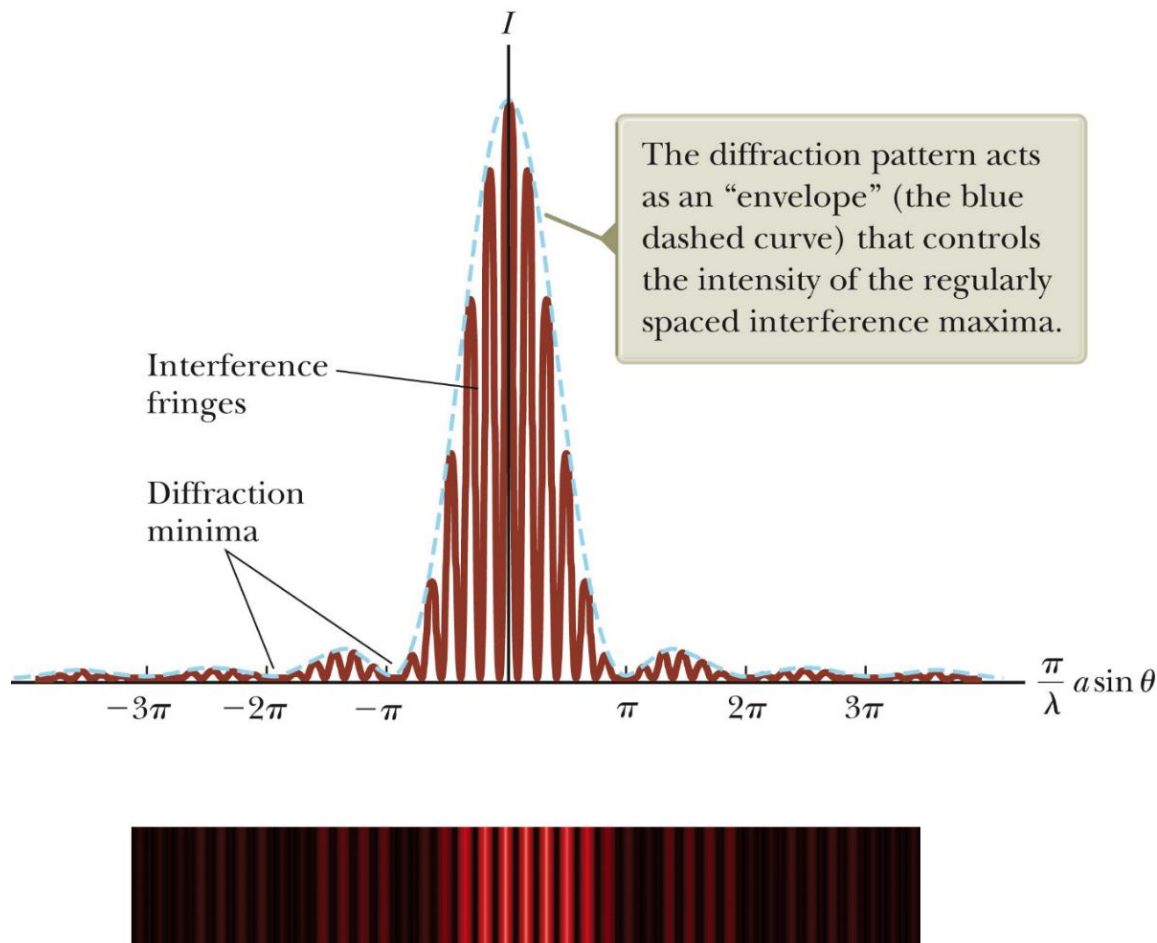
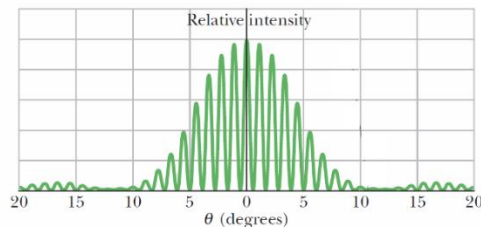
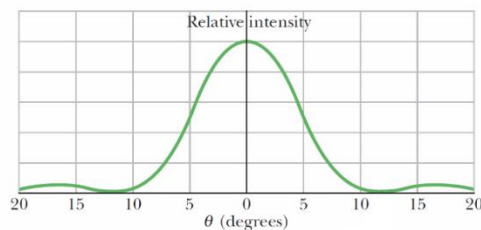
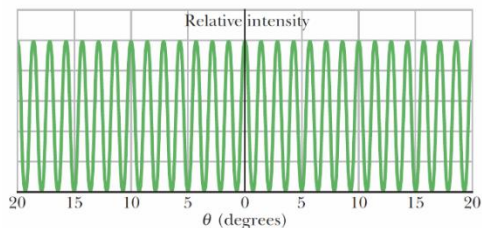


Diffraction by a double slit

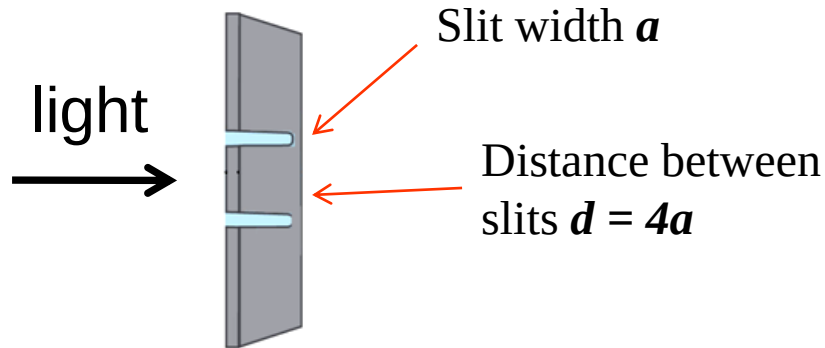


Diffraction by a double slit

In the double-slit experiment of Chapter 35, we did **NOT** consider **diffraction**. Now we consider both diffraction and interference.



A realistic case of two-slit interference pattern



Assume very narrow slit, we have interference pattern :

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \text{ Bright fringes}$$

$$\sin \theta = m\lambda / 4a$$

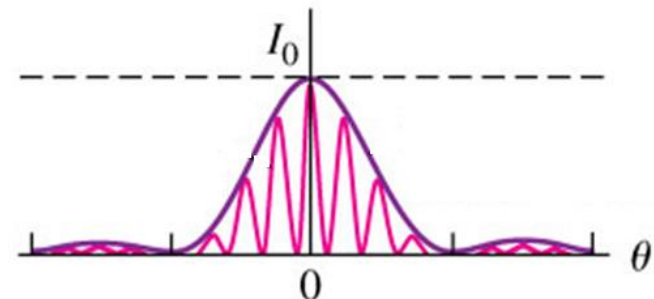
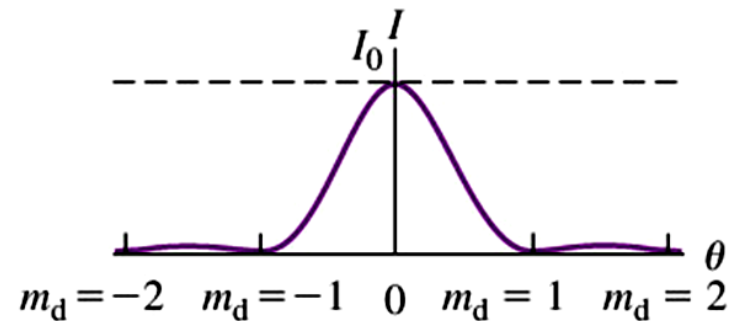
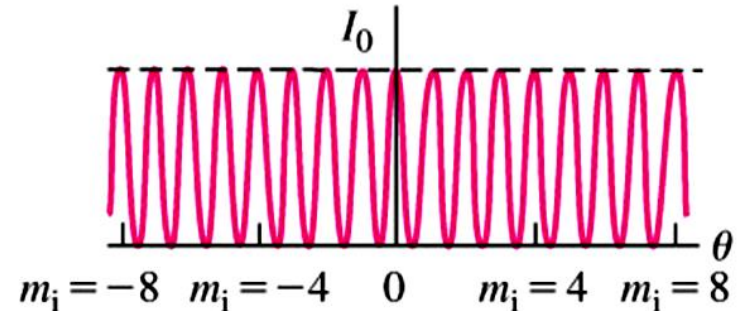
In reality, the slit width a will have a diffraction pattern:

$$\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \text{ Dark fringes}$$

For intensity, we combine the two :

$$I = I_0 \cos^2 \frac{\phi}{2} \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2 \quad (\text{two slits of finite width})$$

$$\phi = \frac{2\pi d}{\lambda} \sin \theta \quad \beta = \frac{2\pi a}{\lambda} \sin \theta$$



Diffraction by a double slit

With diffraction effect, the maxima for the double-slit interference do not have the same intensity. Rather, their intensity change according to a *single-slit diffraction pattern*, i.e., the diffraction pattern acts as an *envelope* over the double-slit interference pattern.

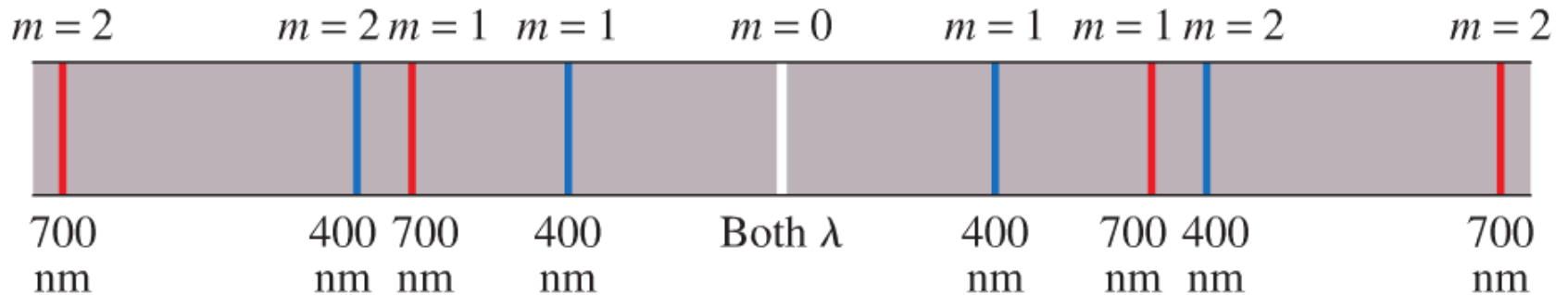
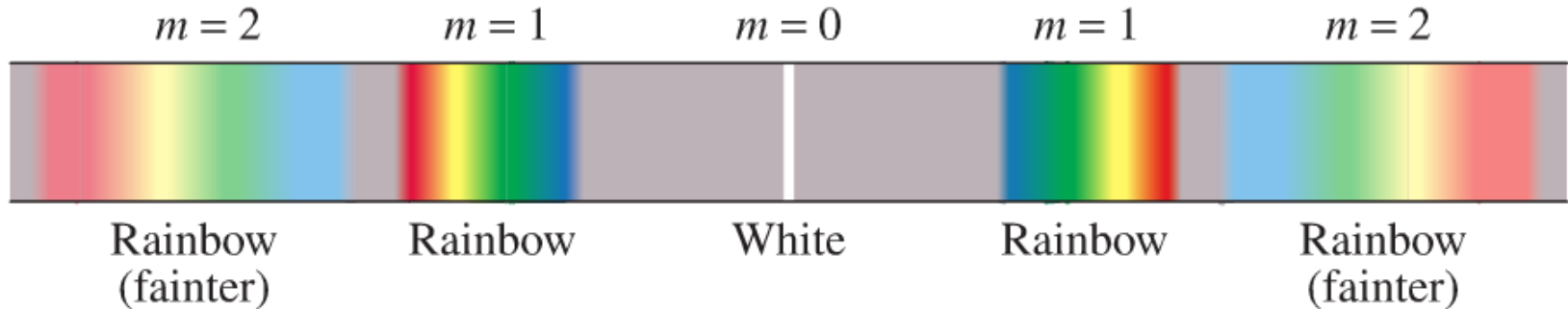
$$I(\theta) = I_m (\cos^2 \beta) \left(\frac{\sin \alpha}{\alpha} \right)^2,$$

in which

$$\alpha = \frac{\pi a}{\lambda} \sin \theta, \quad a: \text{width of each slit}$$

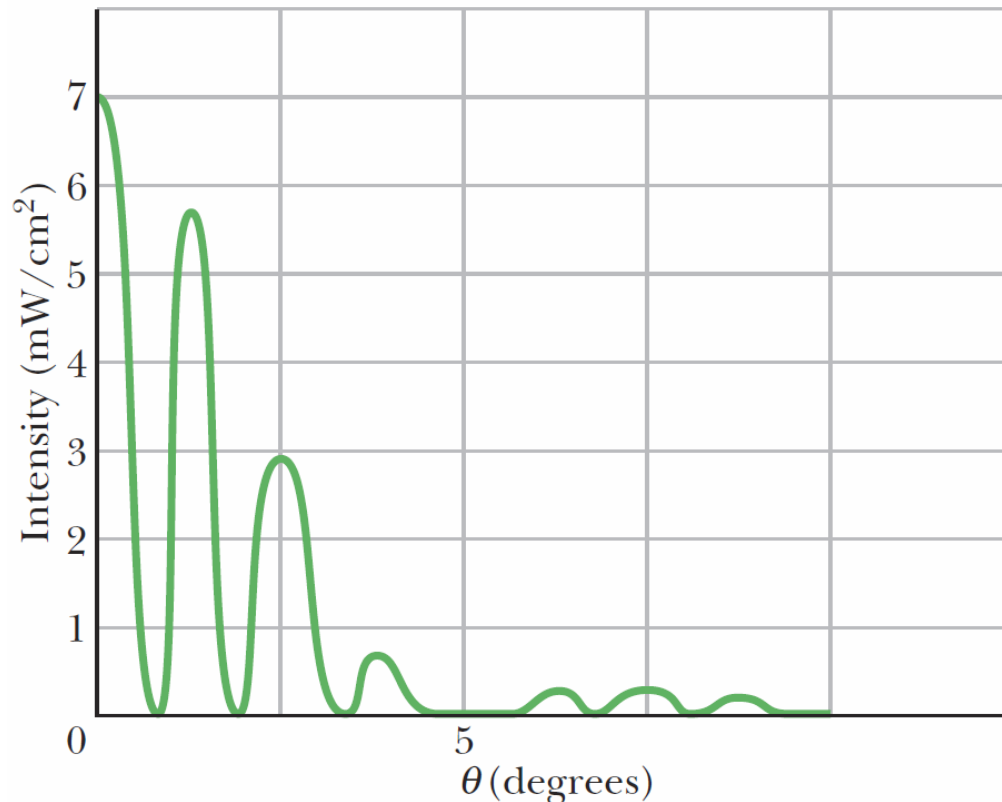
$$\beta = \frac{\pi d}{\lambda} \sin \theta. \quad d: \text{separation of the two slits}$$

Interference of a realistic double-slit using white light



Example

Light of wavelength 440 nm passes through a double slit, yielding a diffraction pattern whose graph of intensity I versus angular position θ is shown in the figure. Calculate (a) the slit width and (b) the slit separation. (c) Verify the displayed intensities of the $n = 1$ and $n = 2$ interference fringes.



Example

Solution:

It is clear from the figure that the $n = 4$ interference bright fringe coincides with the $m = 1$ diffraction dark fringe at $\theta = 5^\circ$.

(a) The slit width a is calculated using the diffraction envelope:

$$a \sin \theta = m\lambda \quad \Rightarrow \quad a = \frac{m\lambda}{\sin \theta} = \frac{(1)(440 \text{ nm})}{\sin 5^\circ} = 5.05 \text{ } \mu\text{m}.$$

(b) The slit separation d is calculated using the interference pattern:

$$d \sin \theta = n\lambda \quad \Rightarrow \quad d = \frac{n\lambda}{\sin \theta} = \frac{(4)(440 \text{ nm})}{\sin 5^\circ} = 20.2 \text{ } \mu\text{m}.$$

Therefore **$a/d = 1/4$** .

Example

Solution (con'd):

(c) We use the intensity distribution equation to get the intensity at $n = 1$ and 2 interference fringes,

$$I(\theta) = I_0 (\cos^2 \beta) \left(\frac{\sin \alpha}{\alpha} \right)^2.$$

For $n = 1$ and 2 ($\cos^2 \beta = 1$):

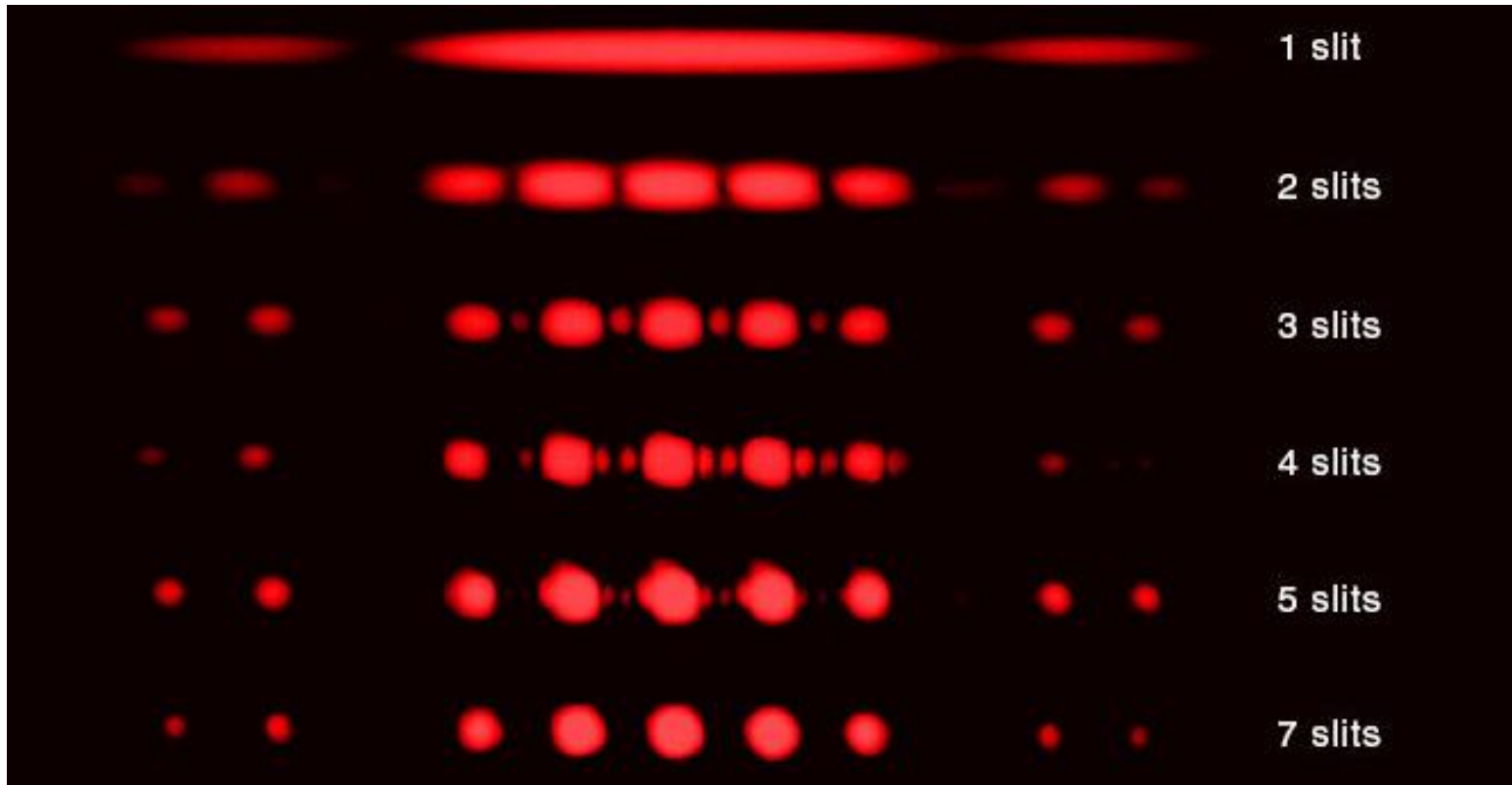
$$\alpha_1 = \frac{\pi a}{\lambda} \sin \theta_1 = \frac{\pi a}{\lambda} \frac{\lambda}{d} = \frac{\pi}{4}, \quad \alpha_2 = \frac{\pi a}{\lambda} \sin \theta_2 = \frac{\pi a}{\lambda} \frac{2\lambda}{d} = \frac{\pi}{2}.$$

so
$$I(\theta_1) = I_0 \left(\frac{\sin \alpha_1}{\alpha_1} \right)^2 = (7 \text{ mW/cm}^2) \left(\frac{\sin \pi/4}{\pi/4} \right)^2 = 5.67 \text{ mW/cm}^2,$$

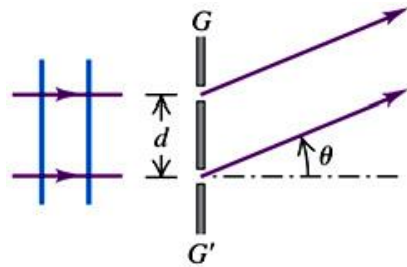
$$I(\theta_2) = I_0 \left(\frac{\sin \alpha_2}{\alpha_2} \right)^2 = (7 \text{ mW/cm}^2) \left(\frac{\sin \pi/2}{\pi/2} \right)^2 = 2.84 \text{ mW/cm}^2.$$

NOT included in the final exam!

Diffraction gratings*



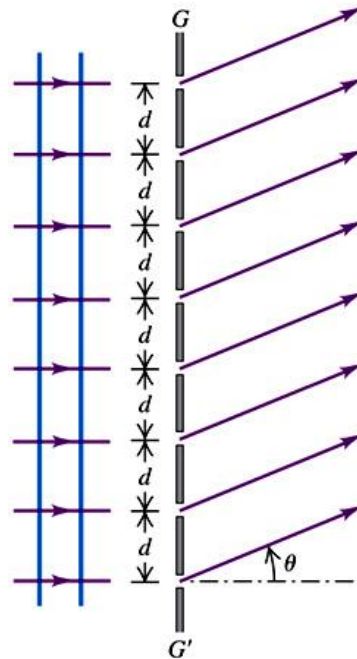
Narrow slits, but many



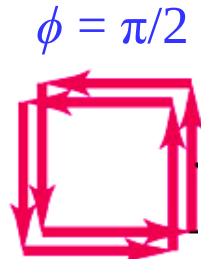
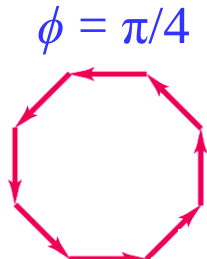
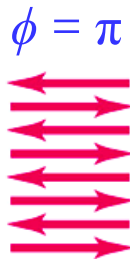
$$d \sin \theta = m\lambda \quad \frac{\phi}{2\pi} \lambda = d \sin \theta$$

ϕ : phase difference between E-fields of light from the two slits

There is only 1 minimum (when $\phi = (2m+1)\pi$) between two consecutive maxima (m and $m+1$).



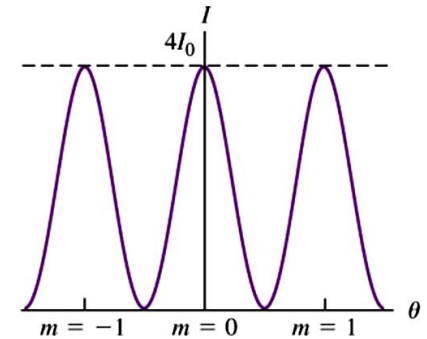
For 8 slits, we can have more than 1 dark fringes between two consecutive maxima:



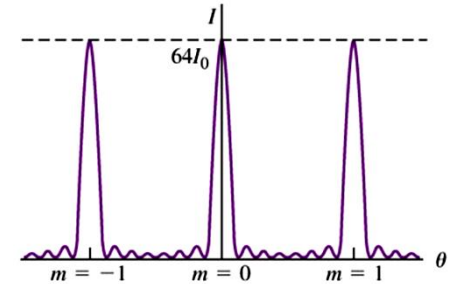
Similar for $\phi = 3\pi/4, 5\pi/4, 3\pi/2$ and $7\pi/4$, so

We have 7 minima for 8 slits!

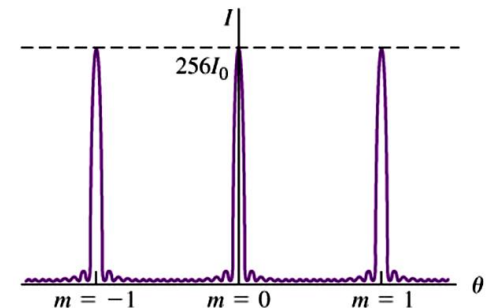
$N = 2$



$N = 8$



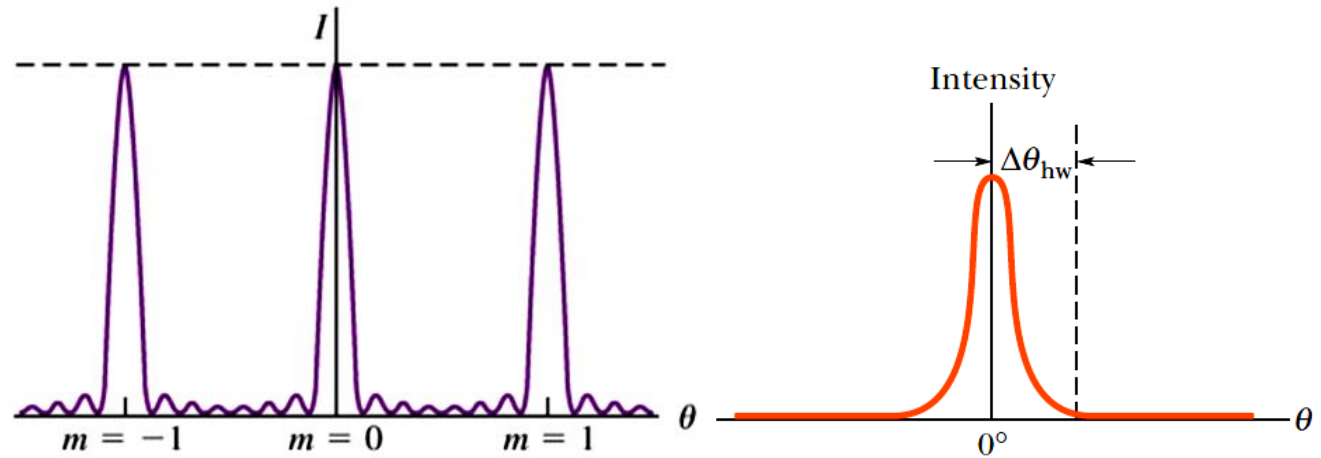
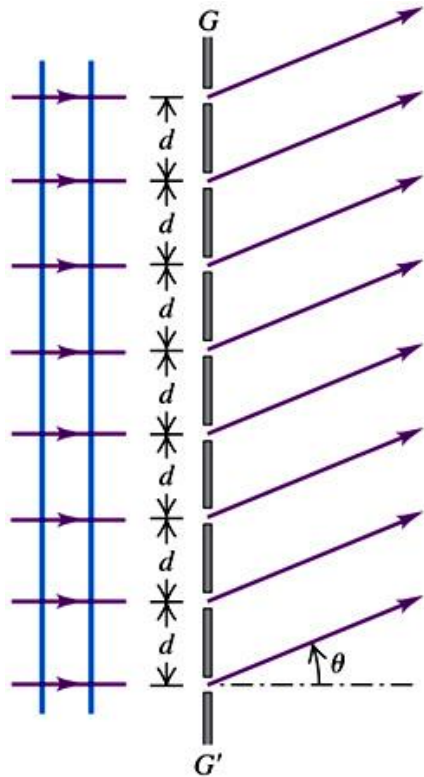
$N = 16$



In general, if you have N slits, there are $N - 1$ minima between each maximum

Intensity of maximum $\propto N^2$

Diffraction gratings



$$Nd \sin \Delta\theta_{\text{hw}} = \lambda \quad \text{Why?}$$

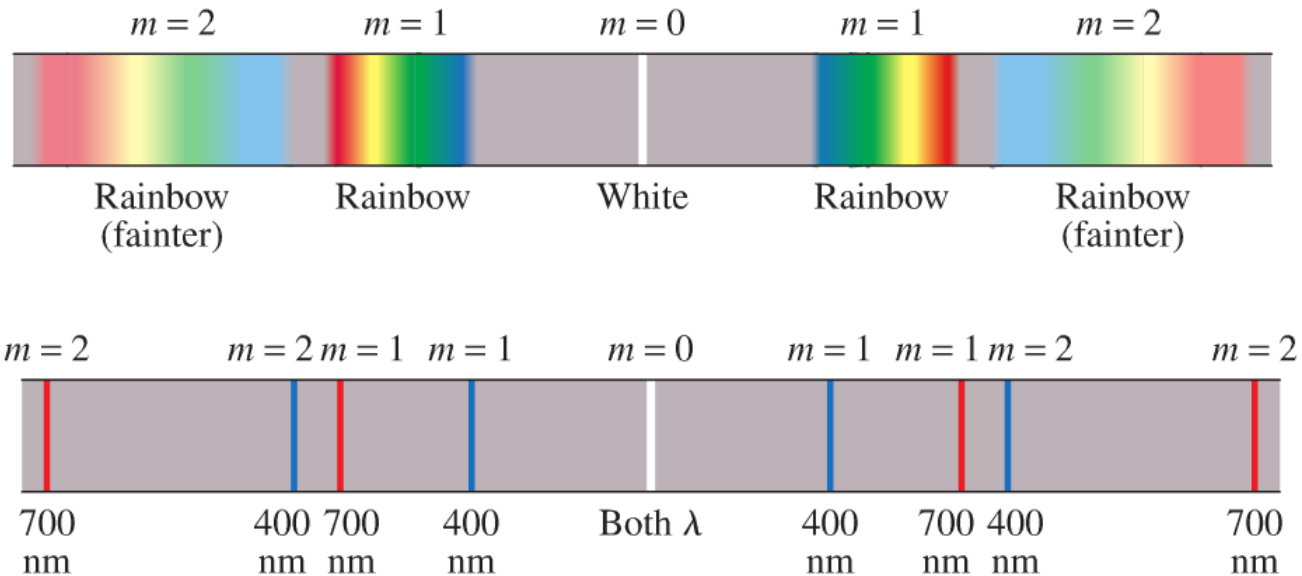
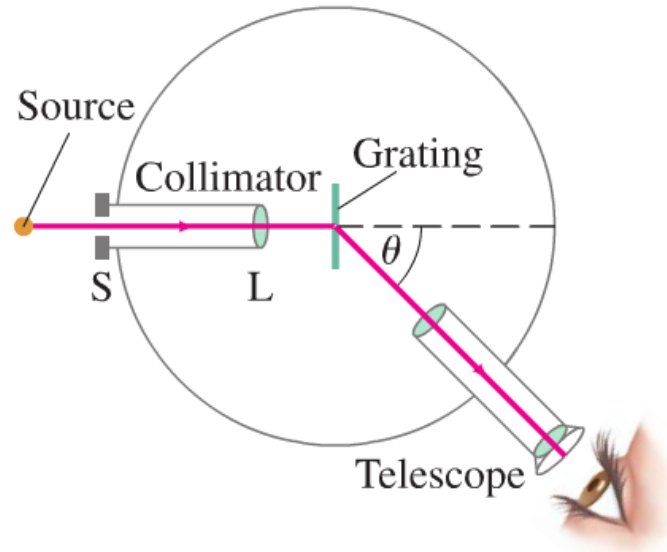
$$\Delta\theta_{\text{hw}} = \frac{\lambda}{Nd} \quad (\text{half-width of central line})$$

Half width of other line at any other locations:

$$\Delta\theta_{\text{hw}} = \frac{\lambda}{Nd \cos \theta} \quad (\text{half-width of line at } \theta) \quad \text{Prove it!}$$

The more slits, the finer the peaks!

Diffraction gratings



Three monochromatic light beams are directed at a diffraction grating. The resulting pattern, shown in greyscale below, is observed on a screen 2 m from the grating. What is the correct order from top to bottom of the three light beams used?

a) green, red, blue

b) red, green, blue

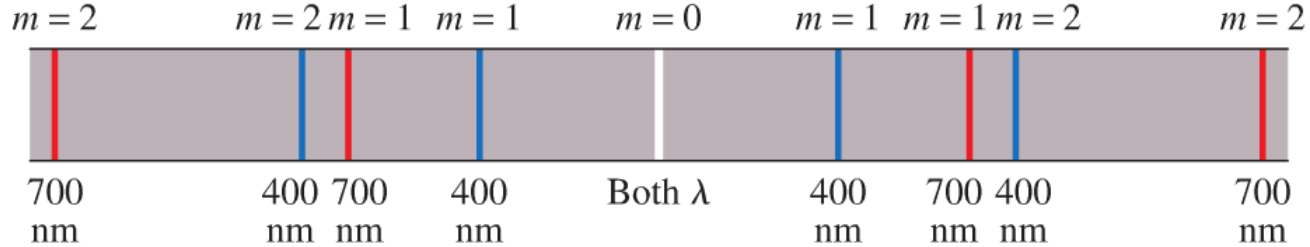
c) blue, green, red

d) green, blue, red

e) blue, red, green



Diffraction gratings : Dispersion & resolving power



Dispersion (D): The angular separation of two lines whose wavelengths differ by a certain amount

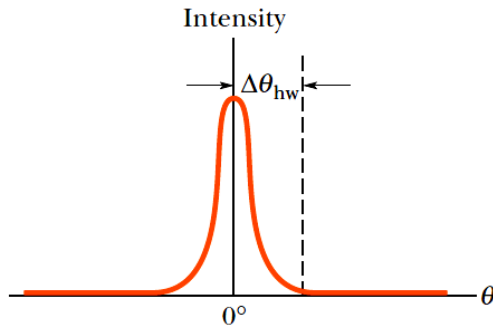
$$D = \frac{\Delta\theta}{\Delta\lambda}$$

$$d \sin\theta = m \lambda \quad d(\cos \theta) d\theta = m d\lambda$$

$$d(\cos \theta) \Delta\theta = m \Delta\lambda$$

$$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos \theta}$$

Resolving power (R): minimum wavelength difference $\Delta\lambda$ that can be distinguished by the gratings



$$\Delta\theta_{hw} = \frac{\lambda}{Nd \cos \theta}$$

Combine the two equations:
$$\frac{\lambda}{Nd \cos \theta} = \frac{\Delta\lambda m}{d \cos \theta}$$

Resolving power **R**:
$$\frac{\lambda}{\Delta\lambda} = Nm$$

The larger the number of slits, the higher the resolving power R

Diffraction gratings : Dispersion & resolving power

Grating	N	d (nm)	θ	D ($^{\circ}/\mu\text{m}$)	R
<i>A</i>	10 000	2540	13.4°	23.2	10 000
<i>B</i>	20 000	2540	13.4°	23.2	20 000
<i>C</i>	10 000	1360	25.5°	46.3	10 000

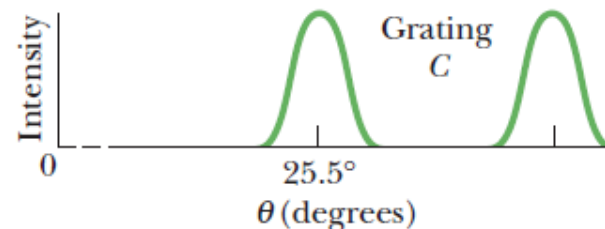
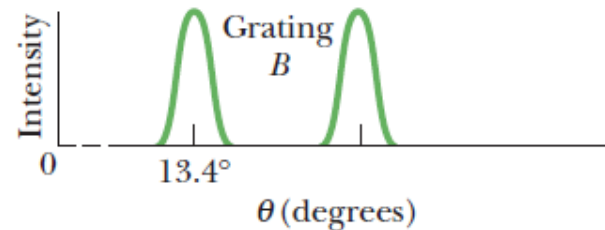
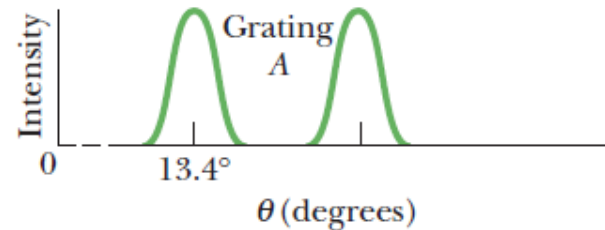
^aData are for $\lambda = 589$ nm and $m = 1$.

Dispersion (D)

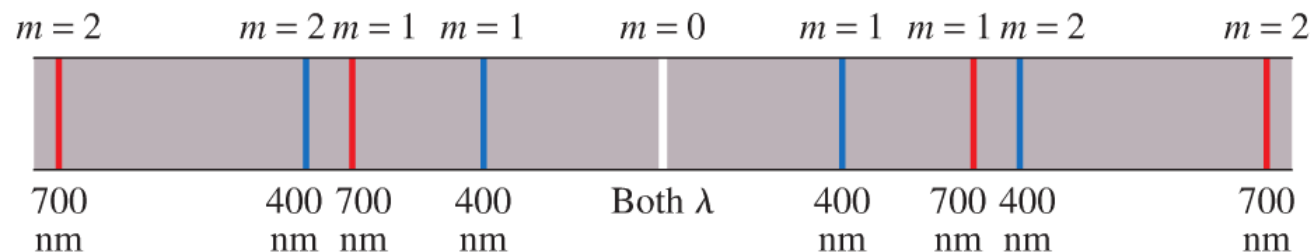
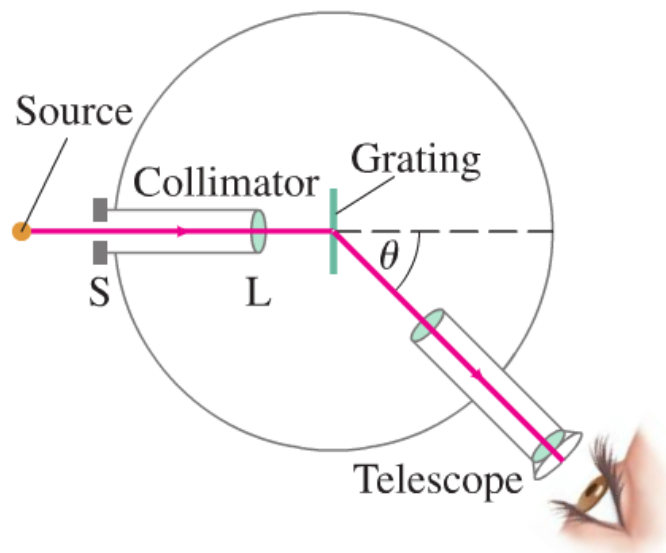
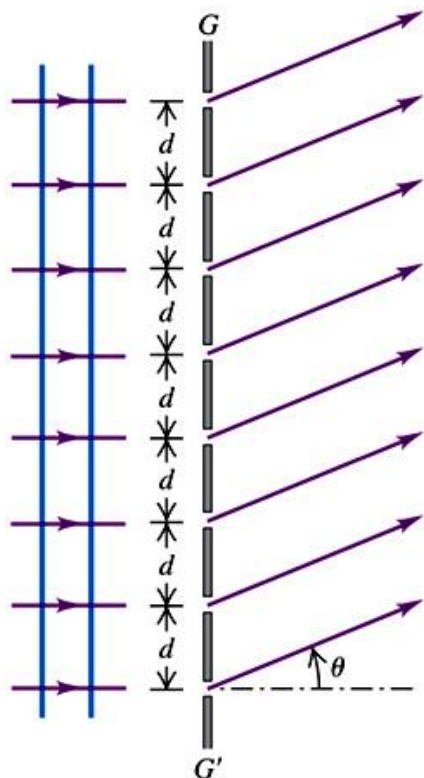
$$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos \theta}$$

Resolving power (R)

$$\frac{\lambda}{\Delta\lambda} = Nm$$



Why bother?



Why do we care about diffraction gratings?

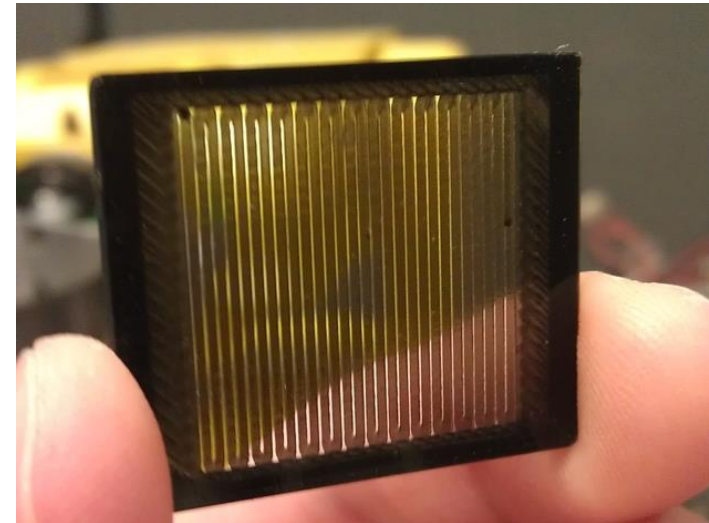
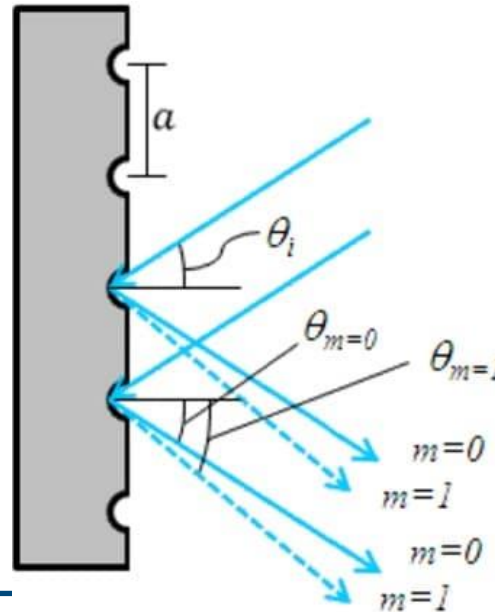
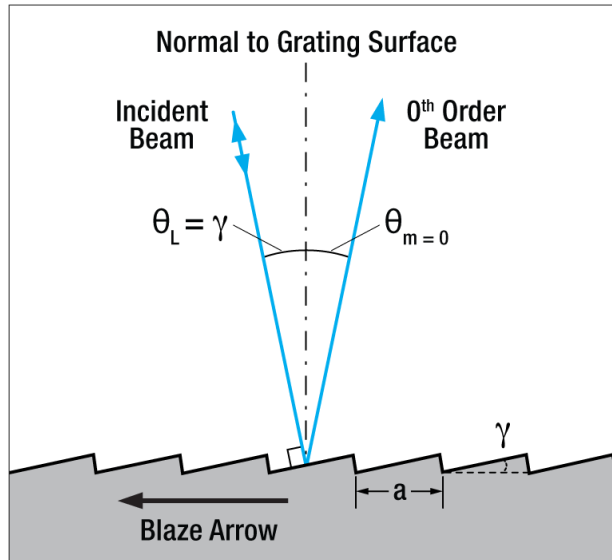
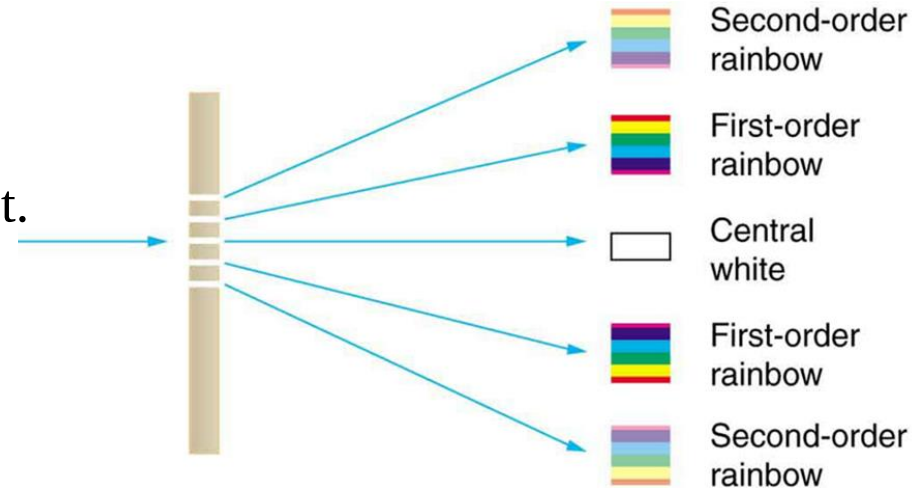
$$d \sin \theta = m \lambda$$

1. With known d , we can measure λ
2. With known λ , we can measure d

Diffraction gratings

There are basically two types of diffraction gratings – that the light passes through a series of slit openings, and that the light reflects from a series of steps at an angle w.r.t. normal reflection.

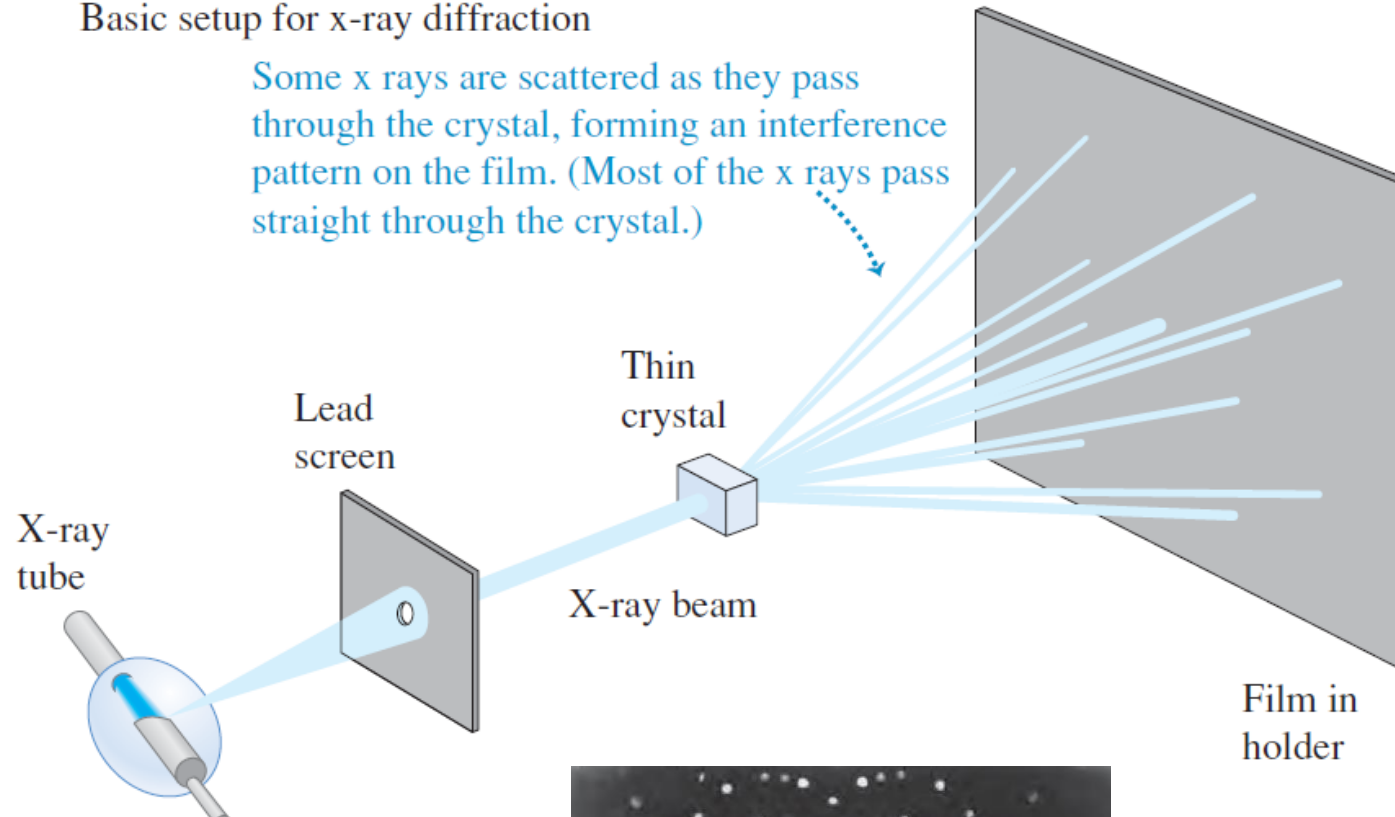
Since the peak width is narrow, color separation is very good. So a diffraction grating is usually used as a monochromator.



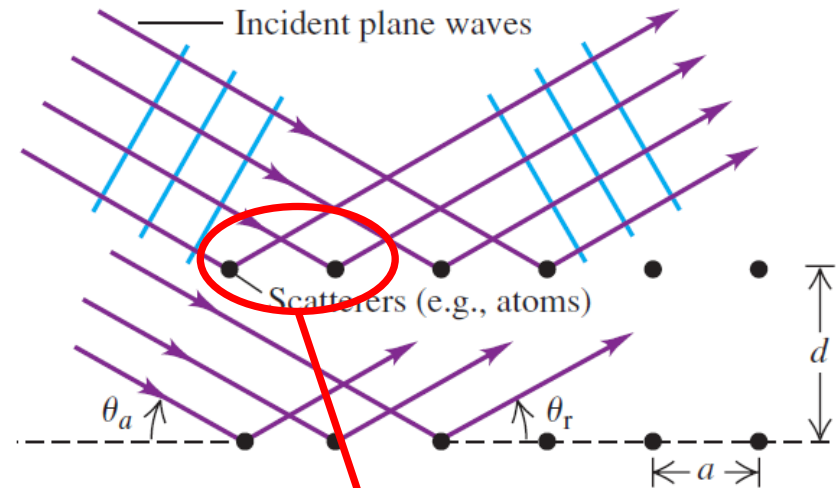
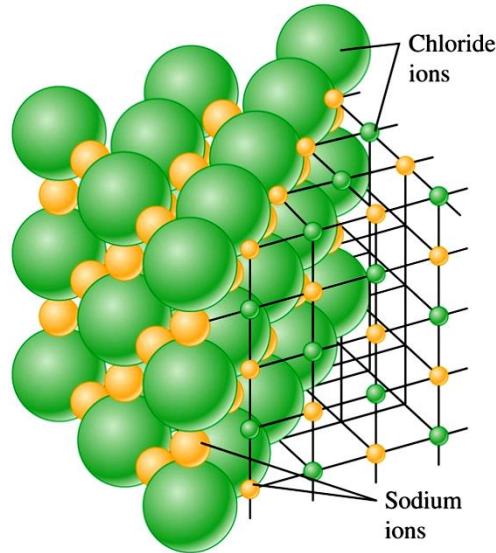
Applications: X-ray diffraction

Basic setup for x-ray diffraction

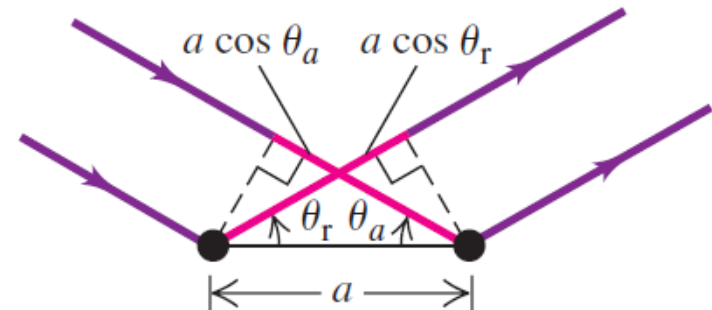
Some x rays are scattered as they pass through the crystal, forming an interference pattern on the film. (Most of the x rays pass straight through the crystal.)



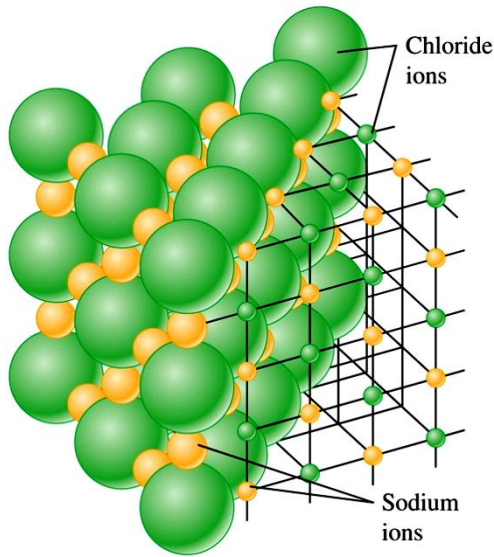
Applications : X-ray diffraction



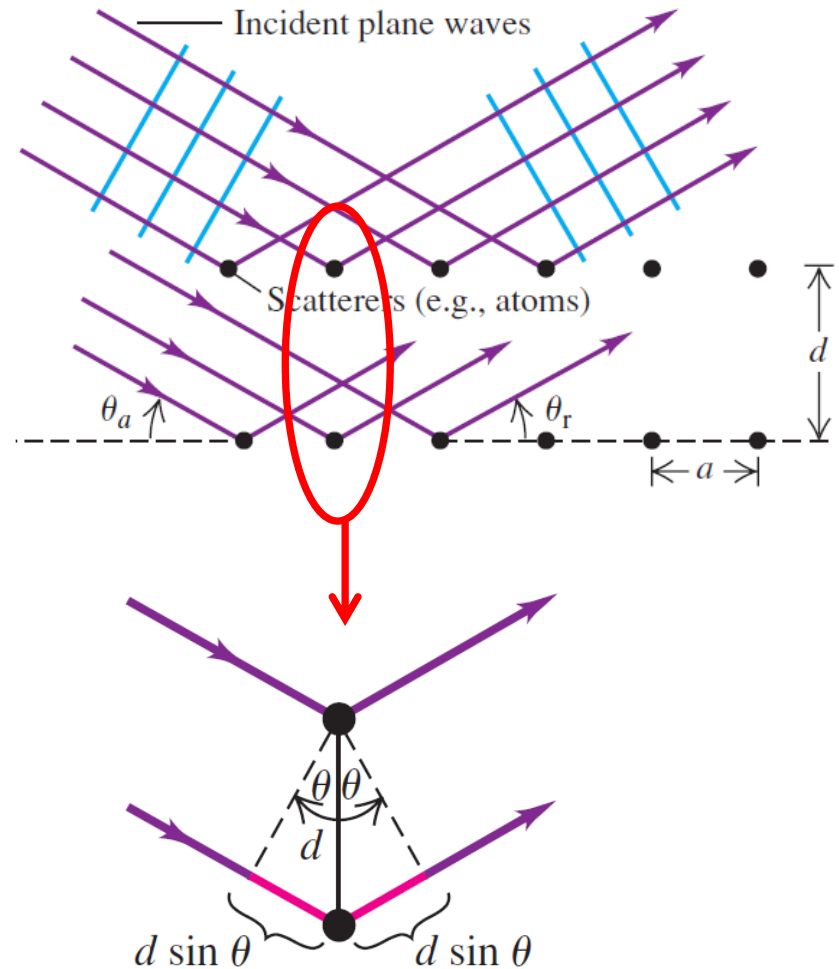
Scattering from adjacent atoms in a row
Interference from adjacent atoms in a row is constructive when the path lengths $a \cos \theta_a$ and $a \cos \theta_r$ are equal, so that the angle of incidence θ_a equals the angle of reflection (scattering) θ_r .



Applications : X-ray diffraction



Scattering from atoms in adjacent rows
Interference from atoms in adjacent rows is constructive when the path difference $2d \sin \theta$ is an integral number of wavelengths.



Bragg's law

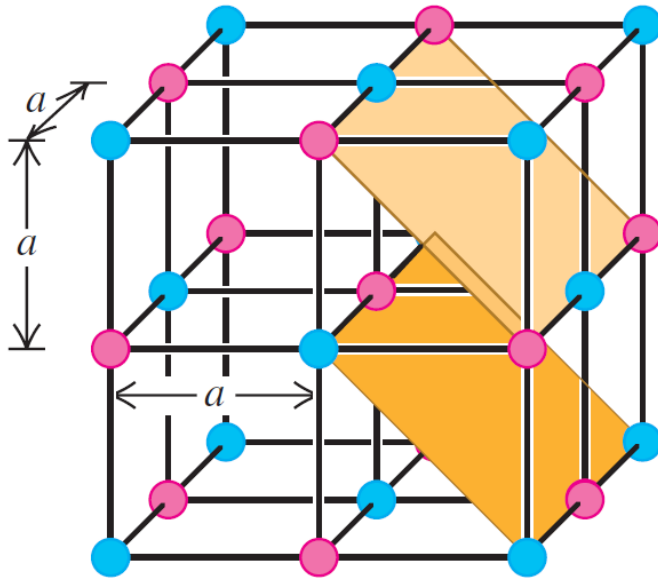
$$2d \sin \theta = m\lambda, \quad \text{for } m = 1, 2, 3, \dots$$

Applications : X-ray diffraction

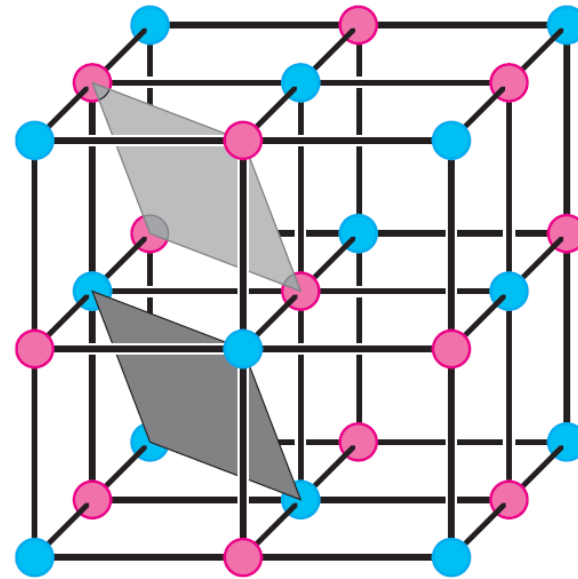
$$2d \sin \theta = m\lambda, \quad \text{for } m = 1, 2, 3, \dots$$

We need to use x-ray since its wavelength is close to lattice spacing!

(a) Spacing of planes is $d = a/\sqrt{2}$.

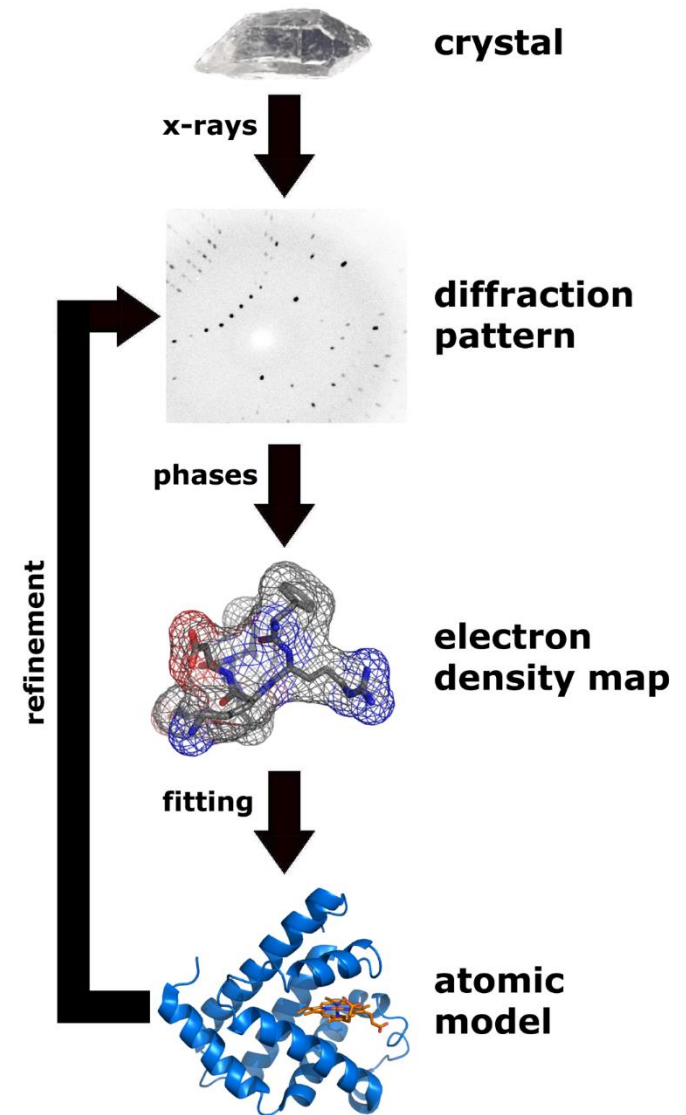


(b) Spacing of planes is $d = a/\sqrt{3}$.



X-ray crystallography

The technique of x-ray diffraction is now very common in the study of crystal structure. By measuring the angles and intensities of these diffracted beams, a crystallographer can produce a three-dimensional picture of the density of electrons within the crystal. From this electron density, the mean positions of the atoms in the crystal can be determined, as well as their chemical bonds, their disorder and various other information.



Example

You direct a beam of *0.154-nm x-ray* at certain planes of a silicon crystal. As you increase the angle of incidence of the beam from zero, the first strong interference maximum occurs when the beam makes an angle of 34.5° with the planes.

(a) How far apart are the planes?

(b) Will you find other interference maxima from these planes at greater angles of incidence?

$$2d \sin \theta = m\lambda, \quad \text{for } m = 1, 2, 3, \dots$$

$$d = \frac{m\lambda}{2 \sin \theta} = \frac{(1)(0.154 \text{ nm})}{2 \sin 34.5^\circ} = 0.136 \text{ nm}$$

$$\sin \theta = \frac{m\lambda}{2d} = m \frac{0.154 \text{ nm}}{2(0.136 \text{ nm})} = m(0.566)$$

Values of m of 2 or greater give values of $\sin \theta$ greater than unity, which is impossible. Hence there are *no* other angles for interference maxima for this particular set of crystal planes.

Chapter 36 Diffraction **Summary**

Review & Summary

Diffraction When waves encounter an edge, an obstacle, or an aperture the size of which is comparable to the wavelength of the waves, those waves spread out as they travel and, as a result, undergo interference. This is called **diffraction**.

Single-Slit Diffraction Waves passing through a long narrow slit of width a produce, on a viewing screen, a **single-slit diffraction pattern** that includes a central maximum and other maxima, separated by minima located at angles θ to the central axis that satisfy

$$a \sin \theta = m\lambda, \quad \text{for } m = 1, 2, 3, \dots \text{ (minima).} \quad (36-3)$$

The intensity of the diffraction pattern at any given angle θ is

$$I(\theta) = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2, \quad \text{where } \alpha = \frac{\pi a}{\lambda} \sin \theta \quad (36-5, 36-6)$$

and I_m is the intensity at the center of the pattern.

Circular-Aperture Diffraction Diffraction by a circular aperture or a lens with diameter d produces a central maximum and concentric maxima and minima, with the first minimum at an angle θ given by

$$\sin \theta = 1.22 \frac{\lambda}{d} \quad \text{(first minimum—circular aperture).} \quad (36-12)$$

Rayleigh's Criterion *Rayleigh's criterion* suggests that two objects are on the verge of resolvability if the central diffraction maximum of one is at the first minimum of the other. Their angular separation can then be no less than

$$\theta_R = 1.22 \frac{\lambda}{d} \quad \text{(Rayleigh's criterion),} \quad (36-14)$$

in which d is the diameter of the aperture through which the light passes.

Double-Slit Diffraction Waves passing through two slits, each of width a , whose centers are a distance d apart, display diffraction patterns whose intensity I at angle θ is

$$I(\theta) = I_m (\cos^2 \beta) \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad \text{(double slit),} \quad (36-19)$$

with $\beta = (\pi d / \lambda) \sin \theta$ and α as for single-slit diffraction.

Diffraction Gratings A *diffraction grating* is a series of “slits” used to separate an incident wave into its component wavelengths by separating and displaying their diffraction maxima. Diffraction by N (multiple) slits results in maxima (lines) at angles θ such that

$$d \sin \theta = m\lambda, \quad \text{for } m = 0, 1, 2, \dots \text{ (maxima),} \quad (36-25)$$

with the **half-widths** of the lines given by

$$\Delta \theta_{\text{hw}} = \frac{\lambda}{Nd \cos \theta} \quad \text{(half-widths).} \quad (36-28)$$

The dispersion D and resolving power R are given by

$$D = \frac{\Delta \theta}{\Delta \lambda} = \frac{m}{d \cos \theta} \quad (36-29, 36-30)$$

and

$$R = \frac{\lambda_{\text{avg}}}{\Delta \lambda} = Nm. \quad (36-31, 36-32)$$

X-Ray Diffraction The regular array of atoms in a crystal is a three-dimensional diffraction grating for short-wavelength waves such as x rays. For analysis purposes, the atoms can be visualized as being arranged in planes with characteristic interplanar spacing d . Diffraction maxima (due to constructive interference) occur if the incident direction of the wave, measured from the surfaces of these planes, and the wavelength λ of the radiation satisfy **Bragg's law**:

$$2d \sin \theta = m\lambda, \quad \text{for } m = 1, 2, 3, \dots \text{ (Bragg's law).} \quad (36-34)$$